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The Measured Voice

Building a voice you can measure

SUEDE SING

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The Measured Voice

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PART

How the voice works

What your voice actually is

Your voice is not an instrument you own. It is one you assemble, freshly, every time you make a sound. Nothing about it is stored. There is no reed, no string, no fixed tube of brass — just moving air, a pair of small folds of tissue that get in its way, and a shaped space above them that colours whatever comes out. That is the whole machine.

Reductive, but it tells you where to look when something goes wrong. Almost every vocal problem you will meet lives in one of those three places, and the three behave nothing alike. Confusing them is how singers end up pushing harder at a problem that pushing cannot solve.

Power, vibrator, filter.

The power: breath

Air leaves your lungs because the space around them gets smaller. That is it. The diaphragm — a dome of muscle under your lungs, separating chest from gut — contracts and flattens on the way in, pulling air down into you. Your ribs swing out at the same time. On the way out, the diaphragm releases upward, the ribs come back in, and the abdominal wall can help squeeze from below.

Here is the part that matters for singing: the diaphragm is an inhaling muscle, so you cannot push with it, and the familiar instruction to sing from your diaphragm is a metaphor rather than anatomy.

What the folds need is pressure below them: air held at a steady, moderate push, arriving at a consistent rate for as long as the phrase lasts. Singers call that *support*. Not force. Pressure. The difference is easy to feel and hard to describe, so try it. Hum a comfortable note and hold it for eight seconds, aiming to make the last second sound exactly like the first. What you have to do to keep it even is a slow, resisted release — ribs staying open a little longer than they want to, abdominal wall coming in gradually rather than collapsing. That resistance is the job. It is a management task, not a strength task.

Which is why more air is so often the wrong answer to a struggling note. Blow harder and you do not get a better note. You get a louder, more pressed one that tires sooner. Chapter four takes this apart properly. For now, hold one idea: support means *steady*, and steady usually means less than you think.

The vibrator: the folds

At the top of your windpipe sits the larynx — the lump you can feel move when you swallow. Inside it, running front to back, are two small folds of tissue. People say "cords," which is a poor image; they are not strings under tension so much as layered flaps, muscle underneath, a flexible layer above that, a thin skin on top.

Breathe normally and the folds sit apart, letting air through without a sound. To make a sound, you bring them together. The air below builds up behind them, pushes them apart, escapes in a puff — and then the folds snap shut again, partly from their own elasticity, partly from the drop in pressure as the air rushes past. Then the pressure rebuilds and it happens again. And again. Sing the A above middle C and that cycle repeats 440 times a second.

So the sound is not the folds humming. It is a rapid chain of air puffs — a buzz. Heard on its own, it would be thin and unmusical, closer to a kazoo than to a singer.

Pitch comes from how fast the cycle repeats, which depends on how long, how thick and how tense the folds are. Stretch them longer and thinner and they cycle faster: higher note. Let them shorten and thicken and they cycle slower: lower note. Different combinations of length, thickness and closure give you the different registers, which is chapter two's territory. Loudness comes mostly from pressure below and from how firmly the folds close on each cycle.

Two things worth noticing. First, the folds are small — a couple of centimetres of tissue at most — and they are managing hundreds of collisions a second. They are built for it. They are also tissue, and tissue does not enjoy being slammed together hard for hours. Second, they work best when the air arriving is steady. Erratic pressure produces erratic tone, which is why breath problems so often show up as pitch problems.

The filter: the tract

Above the larynx is a tube that runs up the throat, through the mouth, and out, with a branch into the nose. This is the vocal tract, and it is where the buzz becomes a voice.

Every tube has frequencies it likes. Blow across a bottle and you get a note; fill it halfway and you get a different one, because the air column changed shape. Your vocal tract does the same, except that you can change its shape while you are singing — lowering or raising the larynx, moving the tongue, opening the jaw, rounding or spreading the lips, letting the soft palate open or close the nasal branch.

The buzz from your folds contains not just the pitch you are singing but a whole stack of higher frequencies above it. The tract lets some of those through loudly and damps others, and which ones get boosted is decided by the shape you have made. Boost one set and the ear hears "ee." Boost a different set and it hears "oh." Same pitch, same folds, same breath — different tube, different vowel.

That is the source-filter idea, and the rest of this book leans on it constantly. The source makes a raw buzz with a pitch. The filter shapes that buzz into a recognisable, coloured sound. They are largely independent: you can sing one vowel across a whole octave, or run through every vowel on one note.

Independent, but not unrelated. Around your break, the resonances of the tract and the frequencies coming from your folds can collide, and a small vowel adjustment can be the difference between a note that clangs and one that rings — that is chapter three. A trained voice carries over an orchestra without a microphone because of a particular band of boosted frequencies produced by tract shape, not by extra effort. That is chapter five.

Why this model earns its keep

Record yourself in the app, hear something you do not like, and this gives you three questions instead of one vague dissatisfaction.

Is it the breath? Wobbling volume, notes that sag at the end of a phrase, running out of air three-quarters of the way through, a tone that goes hard and pressed as you get louder — those usually start below the larynx.

Is it the folds? Breathiness, a note that will not start cleanly, a crack at a predictable spot, a tone that thins out on top — that is generally about how the folds close and stretch.

Is it the tract? Dullness, a woolly vowel, a nasal quality you did not want, a note that suddenly sounds two sizes smaller on one particular vowel — that is shape, and shape is often the fastest thing to change.

Most singers treat every problem as a breath problem and answer it with more effort. Effort is the least specific tool you have. Knowing there are three systems, and that they fail in three recognisable ways, is what turns practice from trying harder into changing something.

One last note on the apparatus. None of these parts were designed for singing. They are a bellows built for oxygen, a valve built to keep food out of your lungs and to seal your chest when you lift something heavy, and a tube built for chewing and swallowing. Singing is a borrowed use of survival equipment. That is why it takes training to do reliably, and why it is worth being careful with. Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more.

Try this

Open the pitch studio, pick a comfortable note in the middle of your range, and hold it for eight seconds on "ah" — then do it again on "ee" without changing anything else. The trace stays on the same line while the sound changes completely. That is the filter working independently of the source.

Registers, and why they exist

Ask ten singers which register they were in on a given note and you will get several answers, each delivered with confidence. That is not carelessness. The words arrived from different rooms — opera studios, church choirs, tracking booths, voice clinics — and nobody ever sat down to reconcile them.

The sounds themselves are not vague. Something specific changes in your larynx as you climb, and it changes in ways you can feel and hear. Start there, and the vocabulary stops mattering so much.

What actually changes

Your vocal folds are not a fixed object. They are layered — a muscular body under a softer cover — and the muscles around them can adjust length, thickness and stiffness somewhat independently of each other. That is registers in one sentence. A register is a way of setting up the folds, and each setup has a band of pitches it handles well and a point where it gives out.

Two muscle groups do most of the negotiating. One runs inside the folds and shortens and thickens them. The other tilts a cartilage in the larynx and draws them longer and thinner. Every note you sing is some balance of those two, plus how firmly the folds are brought together and how much air pressure is arriving from below.

Thick, short folds vibrate through most of their depth. They stay shut for a large share of each cycle and release the air in firm pulses, which builds a tone dense with upper harmonics: solid, carrying, close to speech. Thin, stretched folds vibrate mostly in their cover. Less mass moves, the folds spend less of each cycle closed, and the tone comes out purer and lighter, with less power for the same effort.

Neither is better. They are different tools, and they exist for a plain mechanical reason: no single setup covers a low G and a high B flat. Going up means stretching and thinning, or the folds cannot cycle fast enough without punishing pressure behind them. Going down and heavier means shortening and thickening. The break you feel is the seam between two tools.

The names, and what they point at

Chest voice is the thick-fold setup, the one you speak in. The name comes from sensation: sing a low note with a hand on your sternum and you feel it buzz. The buzz is real, but it is your body resonating sympathetically, not sound being made there. All of it happens at the larynx. Chest voice feels solid, carries consonants well, and turns effortful and shouty when pushed past its comfortable ceiling.

Falsetto sits at the far end — folds stretched thin, often without complete closure, so air leaks through even as they vibrate. That leak is why falsetto tends to be breathy, hollow and hard to sing loudly. Most listeners recognise it in a second: the airy top notes running through soul and indie singing.

Head voice is where the disagreement starts. Some teachers use it as another word for falsetto. Others mean something different — a high, stretched setup in which the folds still close firmly, so the tone rings rather than fogs and can be sung at real volume. Under that second definition, head voice and falsetto are the same neighbourhood at different closure. If you have ever held a high note that snapped from foggy to focused without the pitch moving, you have felt the difference from inside. Chapter 10 covers telling them apart on purpose.

Mix is the most contested term of all, partly because it names two things that usually travel together. There is a laryngeal middle ground, where the folds are neither fully thick nor fully thin and the transition can be crossed without a visible seam. And there is a resonance contribution: shaping the vowel and the tract so a lighter fold setup sounds fuller than it is. Most working singers do both at once, which is why nobody agrees on where the word belongs. Treat mix as a description of a result, not a switch you flip.

Whistle sits above all of it and works differently. Only a short section of the fold length vibrates, the tone is thin and flute-like, and it lives above the top of the standard staff. Some voices find it early, some never do, and having it tells you nothing about whether you can sing.

One way past the vocabulary fight is to stop counting sensations and count vibration patterns instead: fry at the bottom, a thick-fold pattern, a thin-fold pattern, whistle at the top. Four ways the folds can move, rather than a dozen ways the results can be described.

Why teachers disagree

Three reasons, all of them ordinary.

The words describe sensation, and sensation varies. Head and chest name where a singer feels the note ring, and two bodies rarely report the same thing about the same pitch.

The words come from different repertoires. A classical tenor's head voice is a specific, cultivated sound. A pop teacher's head voice might mean anything above the break. A choral director inherited a third vocabulary again. All three are internally consistent and mutually contradictory.

And the boundaries really are blurry. Fold setup is a continuum, not a gearbox. You can sing one pitch with more thickness or less, more closure or less, and any label has to cut that continuum somewhere.

None of this makes the terms useless. It makes them local. When a teacher asks for more head voice, they are asking for a change you can feel — usually less weight, more stretch. Take the instruction, leave the taxonomy.

The part that matters for practice

The most damaging thing register vocabulary has done is turn a mechanical fact into a moral one. Chest gets called the real voice. Falsetto gets treated as an apology. Mix gets sold as the thing serious singers have and you do not. Beginners then spend months hauling chest voice up past its ceiling, because switching feels like losing.

That is backwards, and it is the short road to strain. A register is a band of pitch and volume that one fold setup handles well. Working inside those bands is not a compromise; it is how the instrument is built. Nearly every singer you could name crosses several setups inside a single song, and the skill on display is not avoiding the switch. It is making the switch inaudible — or making it loud on purpose, the way country and yodelling traditions have done for generations.

One boundary is worth stating plainly, because the rest of this chapter is about pushing into unfamiliar territory. Effort is fine. Pain is not. Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more.

For your own practice, the useful questions are mechanical ones. Where does the change happen for you? What does it cost you in volume and tone on either side of it? Can you cross it quietly before you try to cross it loudly? The next chapter takes up that transition zone directly: where it lands, why it lands there, and the principle behind negotiating it.

Try this

Open the pitch studio, pick a comfortable low note, and slide up an octave on an "oo" slowly enough that the pitch trace stays unbroken. Mark the pitch where the tone changes character or the trace starts to wobble. That is your register transition announcing itself, and it is the same spot you come back to in weeks 5 and 6.

The passaggio: where it breaks and why

There is a place in your voice where things stop being easy. You climb a scale, everything is fine, fine, fine, and then one note arrives with a wobble in it, or a crack, or a sudden thinning, or the feeling that you have to shove. Singers call it a wall, a step, a gear change, a hole. The Italian word is *passaggio* — passage — and it beats every English option, because it names a place you go through rather than a thing that stops you.

Almost every voice has one. Trained singers have not removed theirs; they have made the crossing quiet enough that a listener stops noticing it. That is the goal worth aiming at, and it is worth saying early, because a lot of practice time gets spent trying to delete a permanent feature of the instrument.

What is changing

Two muscle groups do most of the pitch work and pull against each other — one thickening the folds, one stretching them, as chapter two sets out. Low, full singing leans on the thickening, high singing leans on the stretching, and every note between is a blend that has to shift as you climb.

That shift is not the problem. The folds change proportion across most of your range without your noticing. The problem is that the shift is not the only thing happening.

Above the folds, the throat and mouth act as a filter — the resonator that turns one buzz into "ah" or "ee". Chapter five works through that properly. What matters here is that the filter has resonant peaks, and for a given vowel and a given shape of throat, those peaks sit at fixed frequencies. Your pitch is not fixed. It is climbing. As it climbs, the harmonics of your voice slide upward through those fixed peaks like a train passing signal lights.

For most of your range this is fine. The harmonics slide past, the tone changes colour a little, nobody minds. But there is a region where a strong harmonic — often the second one, the octave above the note you are singing — arrives at the lowest resonant peak of the vowel, sits on it, and then keeps climbing past it. On the way up to the peak, the voice feels loud and free and reinforced. Past the peak, that reinforcement drops out from under you. The same effort buys a smaller note.

Now put the two things together. Right where the acoustic support goes away, the muscular blend also has to change. If you answer the drop in loudness by pushing harder — the natural human response to a sound getting quieter — you drive the thickening muscles at the exact moment they need to yield. An adjustment that was going to spread over several notes now happens in one. You hear it. That is the crack.

Roughly where it lands, and why

The passaggio sits where it does because it is tied to the dimensions of your instrument. Longer, thicker folds and a longer vocal tract put the whole business lower. Shorter and lighter put it higher. That is why voice type and passaggio location travel together, and why the transition tells you more about the kind of voice you have than the highest note you can hit does.

Teaching traditions generally describe two turning points rather than one: a lower one where the voice has to start shedding weight, and an upper one, often a fourth or fifth above, where the head mechanism takes over properly. The conventional approximate locations:

- **Bass:** around A3, then around D4
- **Baritone:** around B3, then around E4
- **Tenor:** around C4 or D4, then around F♯4 or G4
- **Contralto:** around E4, then around A4
- **Mezzo-soprano:** around F4, then around B♭4
- **Soprano:** around F♯4 or G4, then around C♯5 or D5

Treat those as a neighbourhood, not an address. Individual voices sit a semitone or three off the list in either direction, and the vowel you are singing shifts the acoustic half of it, because different vowels put their resonant peaks in different places. An "ee" and an "ah" on the same pitch do not meet the wall at the same moment.

The pattern is the part you can use. If you find a zone of two to four notes where sustaining feels like work while the notes just below it feel free, that zone is probably yours, and it tends to stay put rather than wander week to week. Chapter ten covers pinning down your own in the pitch studio. For now it is enough to know the thing exists and has a physical address.

Why force is the wrong tool

Here is the trap: pushing works, for about two notes. The heavy, thickened configuration can be dragged higher than it wants to go if you raise the air pressure under it, and the result is loud and exciting and feels powerful. Then it stops. The ceiling arrives without warning, the voice cracks into something thin or refuses outright, and the singer concludes they have found the top of their range.

They have found the top of *that configuration*. There is often more voice above it, reachable by giving something up rather than by adding something. That runs against instinct, and it is a common reason singers plateau in the middle of the voice rather than at the edge of it.

Force also does the reverse of what it promises acoustically. Blasting more air through a heavy fold setup, at a pitch where the resonance has already dropped away, does not bring the resonance back. It raises the collision force on the folds while the sound stays small. More work, less voice. Doing that loudly and often is how singers wear themselves down, which is chapter six's subject and worth reading before you spend much time up here.

One rule overrides everything else in this chapter: singing should not hurt. Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more.

What works: vowel and weight

Two adjustments, both subtractive.

Weight. Let the sound thin as you approach the zone. Not breathier, not collapsing — thinner. A dimmer, not a switch. Sing the approach a little quieter than feels natural, so the muscular handover can happen over several notes instead of as an emergency. Once you are above the zone, the volume can come back. It is the crossing that needs to be quiet.

Vowel. Since part of the trouble is a harmonic arriving at a fixed resonant peak, move the peak. That is all vowel modification is. Shading a bright, narrow vowel toward a rounder, more neutral one changes the shape of the tube, which moves where the reinforcement sits, which lets the harmonic pass without the sudden dropout. Chapter five gives the ladder of which vowel shades toward which. Small amounts. If a listener can hear the word change, you have gone too far.

The two work together. Vowel without weight release still fights the handover. Weight release without the vowel gives you a note that is easy and hollow. Together they spread the transition across more notes, which leaves less of a seam to hear.

Expect it to feel worse before it feels better. Giving up the heavy setup makes the passaggio notes sound smaller from inside your own head, and that is where singers abandon the approach, because quieter feels like worse. It is not worse. It is the version with somewhere to go.

Try this

In the pitch studio, sing a slow siren on "ah" from a comfortable low note up through the zone where things get effortful, at about half your usual volume, letting the vowel round toward "aw" as you climb. Watch the pitch trace for the moment it jumps or thins, and note where that moment sits. Repeat the slide a few times across a few sessions; whether that spot smooths out or moves is the thing worth watching here.

Breath: support versus pressure

Support is usually taught as effort. Push from the belly. Engage the core. Really give it. All of that describes muscles working hard, which is not the same thing as the vocal folds getting what they need.

Here is what they need. Your folds sit closed, or nearly closed, across the top of your windpipe. Air below them is under slight pressure. That pressure blows them apart, they snap back, and the cycle repeats hundreds of times a second. For the tone to stay even, the pressure underneath has to stay even. That is it. Support is the job of keeping that pressure steady while your lungs are emptying, your ribs are collapsing, and the phrase is asking for different notes at different volumes.

Steady is the operative word, not strong. A phrase that starts loud and thins out at the end is a pressure problem. So is a note that wobbles in intensity. A high note that goes sharp and strident is usually too much pressure. None of them is fixed by more air.

What the ribs and diaphragm actually do

The diaphragm is a dome of muscle under your lungs. When it contracts it flattens downward, the space in your chest gets bigger, pressure drops, and air comes in. That is the whole of its job. It is an inhaling muscle. It does not push air out, and it cannot — muscles pull, they do not push, and there is nothing for it to pull on that would drive air upward.

So when a teacher says "support from the diaphragm," what is really happening is that you are letting the diaphragm release slowly instead of letting it snap back. Your ribs work the same way. The muscles between them can hold the ribcage open, and if you keep it open a little longer than it wants to stay open, the air leaves more slowly and more evenly. The abdominal wall comes back in gradually rather than in a lurch.

That is the entire mechanism: two forces in a slow argument. Elastic recoil wants to empty you fast. A gentle inspiratory hold wants to keep you full. Support is the negotiated middle, and the tone tells you whether the negotiation is going well.

You can feel this without singing a note. Sit up, hand on the side of your lower ribs. Breathe in through the nose for four counts and notice the ribs widen sideways — not the shoulders rising, the ribs going out. Then hiss out on a steady "sss" for as long as it stays even, and pay attention to the ribs. If they drop at once, you are dumping. If you can keep them broad for most of the hiss and let them come in only near the end, that is the sensation you want under a sung phrase.

Why more air is the wrong fix

When a note feels weak, the instinct is to send more air at it. It almost never works.

Too much airflow blows the folds apart harder than they can spring back. The closure gets sloppier. Instead of a clean, efficient cycle you get a leaky one, where a lot of air escapes as noise rather than converting to tone. The note gets breathier and quieter for more effort, so you push harder, and now you are working the throat to hold the folds together against a gale you created. That is where singers get tired fast and blame their stamina.

There is a second failure, and it is the noisier one. If you press the abdominal wall in hard while the folds stay firmly closed, pressure climbs fast. Pitch rides on pressure to a degree, so the note goes sharp. Tone gets hard and pinched. Vibrato either flattens out or turns into a shudder. It sounds like commitment. It is squeezing.

The test is simple. Sing a phrase that has been giving you trouble at about half your usual volume, deliberately using less air. Often the pitch settles and the tone rounds out. If it does, the problem was not a shortage of breath.

Actual under-support looks different, and is rarer than people think: the tone starts thin, the pitch sags flat, and the phrase runs out early even though you took a big breath. That is a collapse — you inhaled and then let everything fall in the first second. The fix is not a bigger inhale. It is holding the ribs open past the first beat.

The size of the breath

Most singers take too much air, not too little. A full-to-bursting inhale leaves your ribcage stretched near the top of its range, where elastic recoil is strongest. You are now fighting your own tissue just to keep the first note from blasting, and you spend the phrase getting rid of surplus air rather than using it.

Aim for comfortably full — think three-quarters, not the deepest breath you can manage — and take it in silently. A gasping inhale means air is rushing through a narrow throat, and a throat that is narrow on the way in tends to still be narrow on the first note. Silent means open.

Breathe low and wide rather than high. If your shoulders climb toward your ears on the inhale, you are filling the top of your lungs, which is the least useful part and the one that most often drags tension into the neck. The visible movement should be around the lower ribs and the belly.

What steady pressure feels like

Sensations are personal, and you should be suspicious of anyone who describes them as universal. A few come up often enough to name.

The lower ribs stay broad through most of the phrase, rather than dropping the moment you begin. There is a mild sense of resistance around the waist — not a clench, more like leaning gently against something. The throat is uninvolved. That last one is the real marker: if support is working, your neck feels about the same on a loud note as on a quiet one. If it grips as you get louder, the effort has migrated to the wrong place.

Loud is where this gets tested. Getting louder does mean more airflow and more pressure, but it also means firmer closure at the folds, and the two have to rise together. Add air without the closure and you get breathy and loud. Add closure without the air and you get pressed and tight. The good version feels like the sound is getting bigger without the effort moving upward into the throat.

A correction worth trying on a pressed sound: think of the volume as coming from the ribs staying open longer, not from a harder push. It is a mental relabel more than a mechanical change, and it tends to take the squeeze out anyway.

Practising it away from pitch

Breath work is one of the few things worth practising with no singing attached, because you can watch the pressure directly instead of guessing at it through tone.

Slow exhales on a hiss, a lip trill, or a straw into a glass of water all do the same thing: they put a small resistance in front of the airstream so you have something to meter against. The straw and the lip trill have a further advantage. The back pressure they create makes hard blowing obvious. You cannot blast through a lip trill — it just stops.

Box breathing, the Farinelli drill and the sustain test each train that slow release from a different angle, and chapter fourteen describes all three in full. They are tabs in the Breath section of the app, and they belong early in a practice session rather than at the end, when you are tired and more likely to substitute effort for control. Lip trills live in Warmups.

One caution, and it applies to every held-breath drill here. If you feel light-headed, stop — do not finish the round. Nothing in this chapter should hurt. Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more.

Breath is not fuel you spend, and it is not force you apply. It is a pressure you hold steady underneath a vibration, and the skill is in the holding, not in the amount.

Try this

Open Breath and run the sustain test twice — once cold, then again after a minute of lip trills. Compare both numbers it gives you, seconds and steadiness, not just the seconds. A second attempt that lasts longer at equal or better steadiness suggests you were spending air rather than running out of it. Your last ten attempts stay on the chart, so you can see whether that pattern holds up over a few sessions or was a one-off.

Resonance, placement and the vowel

Your folds make one sound, and it is not a nice one. Heard raw, before it travels anywhere, it is a thin buzzy rasp — closer to a duck call than to singing. Everything you recognise as your voice happens after that buzz leaves the folds. The vowels, the warmth, the edge, the sense that this is you and not someone else: all of it gets added on the way out, in the corridor between your folds and your lips.

That corridor is the vocal tract. It is a tube, and you reshape it constantly with your tongue, jaw, lips, soft palate and larynx height. The buzz is the source. The tube is the filter. Once you have that split, most tone problems stop looking like effort problems and start looking like shape problems.

What a filter actually does

Any tube of air has frequencies it likes. Blow across a bottle and it sings one note; pour water in and the note rises, because you changed the size of the air column, not the blowing. Your tract works the same way, except that it is an irregular tube with several resonant peaks rather than one, and you can move them around mid-phrase.

Those peaks are called formants. The buzz already contains a fundamental plus a long stack of harmonics above it, and the tract adds nothing to that stack. It only decides which parts get amplified and which get damped on the way out. Harmonics that land near a formant come out loud. Harmonics that land between formants come out quiet.

This is why one pitch can produce several completely different sounds. Sing a comfortable note in your middle and slide the vowel from "ee" to "eh" to "ah" to "oh" to "oo" without changing pitch or volume. The note stays exactly where it was. The colour changes enormously. You moved two formants with your tongue and lips and never touched the note.

So a vowel is not a mouth position you memorise. A vowel is a pair of resonant frequencies. Different mouth shapes that land on the same pair produce the same vowel, which is why someone whose face and throat are nothing like yours can say "ah" and you recognise it instantly.

Vowel modification, and why it is not cheating

Low in your range, your fundamental sits well below the first formant of every vowel, with plenty of harmonics packed underneath to feed the resonances. Sing whatever vowel you like down there; it works. As you climb, the harmonics spread apart, and eventually the fundamental itself starts closing in on the first formant of the vowel you are trying to sing.

At that point the vowel fights you. Closed vowels have low first formants, so "ee" and "oo" run out of room earliest. Open vowels like "ah" sit higher and survive further up. Push a vowel past the point where its resonance can support the pitch and the sound thins, goes shouty, or refuses to speak at all. The instinct is to add force. Force is the wrong tool. The tube is the wrong shape.

Vowel modification is the fix. You nudge the vowel toward a neighbour so its resonance moves back into a workable relationship with the pitch. In practice that means:

- Bright, closed vowels open a little as you ascend. "Ee" drifts toward "ih", then toward "eh". "Oo" drifts toward "uh" or "oh".
- Wide open vowels round a little at the very top. "Ah" drifts toward "aw" or "uh", which is why high notes often come out vaguer than the printed word.
- The jaw usually needs to release more than you think, and the tongue usually needs to stay higher and further forward than you think. Two separate levers. Beginners weld them together.

The word "modification" makes it sound like a compromise, a degraded version of the real vowel. It is the opposite. A listener rebuilds the intended word from context far more generously than you expect, and a well-modified high note reads as clearer than a stubbornly literal one that has gone shrill. Plenty of singers who do this could not tell you they were doing it.

Placement is a useful fiction

You will be told to put the sound in your mask, forward, behind the eyes, in the cheekbones, over the soft palate, in a spot two inches in front of your face. None of that is physically true. Sound is not routed anywhere. It radiates out of your mouth and, to a small degree, vibrates the bone and tissue around it. You cannot aim it.

What you can do is notice sensation. When the tract takes a shape that builds strong resonance in one frequency region, you genuinely feel buzzing in particular places — the front of the face for bright, high sounds, the chest and throat for lower ones. The sensation is real. It is a consequence of the tone, not a cause of it.

That makes placement language a shortcut rather than a lie. "Forward" compresses a whole cluster of adjustments — a higher, more forward tongue, a narrower space above the larynx, more high-frequency energy — into one word you can act on in the second before you sing. If a cue produces the sound you want, use it. Just do not build a theory on it, and do not stand there trying to deliver sound to your cheekbones while the tone refuses to change. When a cue stops working, drop it and go back to the levers you can actually move: vowel, tongue, jaw, larynx height, soft palate.

Brightness, and why some voices carry

There is a band of frequencies, well above any pitch you are actually singing, where human hearing is at its sharpest and where a band or an orchestra tends to put comparatively little energy. Trained singers, classical ones especially, often develop a strong resonance peak sitting right in that band. It is usually called the singer's formant, and it goes together with a comfortably low larynx and a narrowing of the tube just above the folds.

The practical consequence is that carrying power is mostly a resonance property rather than a loudness one. A voice with that peak cuts through an ensemble without being pushed. A voice without it can be louder at the mouth and still disappear in the mix. It also explains why singing harder rarely solves a balance problem: volume adds energy across the whole spectrum, including the regions the band already owns.

Contemporary singers get a related effect from twang, narrowing the same region to add a ringing, sometimes nasal-sounding edge. Twang is efficient rather than effortful. It brightens the tone by reshaping the tube, not by driving more air or squeezing harder. It is also easy to overdo and easy to confuse with strain. One rough check: if the brightness costs you nothing and you can still sing the same phrase quietly, it is coming from resonance. If holding it takes gripping, if your throat tightens, or if the sound turns rough, stop and go back to the plainer version — that is force, and force is not something to practise through. Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more.

A pairing worth hearing yourself: a straight, unbrightened tone next to the same note with a slightly narrowed, ringier setup. Same pitch, same air, different tube. That comparison is the whole chapter.

Try this

In the Recorder, sing one sustained note in your comfortable middle three times — once on a plain "ah", once slid toward "aw", once with a brighter, narrower ring — then A/B compare the takes. Listen for which one carries best on playback rather than which one felt biggest while you were singing it.

Stamina, warming up, and not hurting yourself

Your voice is tissue doing a job. Two small folds meeting hundreds of times a second, driven by air, shaped by a tract that also has to swallow and breathe. Tissue that works gets tired. That is not a character flaw or a sign you are doing it wrong — your legs are the same arrangement.

So the useful question is not how to avoid fatigue. It is how to tell which kind you are feeling, and what a session should look like on either side of it.

What a warmup is actually doing

A warmup is not stretching. You cannot stretch a vocal fold the way you stretch a hamstring, and nothing you do in ten minutes makes the tissue longer.

What changes is coordination. Muscle that has been moving behaves differently from muscle that has been still since you woke up. The thin fluid layer over the folds behaves differently too. Most of all, the relationship between breath pressure and fold closure gets recalibrated, and that relationship drifts — overnight, after a silent afternoon, when you are tired. A warmup is mostly the work of finding it again.

Which is why good warmups feel boringly small. Quiet, mid-range, narrow. Lip trills, a hum, an "ng", singing through a straw into water, a five-note pattern that barely leaves the middle of your voice. They share one structural trick: a partial obstruction in front of the folds changes the pressure balance above and below them, so the folds can vibrate fully without you pressing. You get the movement at a fraction of the effort.

The shape is small to large. Start where your voice is already comfortable and quiet. Widen the range before you add volume. Add volume before you add anything sustained or high. Ten to fifteen minutes covers a practice session. The moment it starts to feel like work, you have crossed out of the warmup and into training — fine, but be honest about which one you are doing.

Two things a warmup does not do. It does not make you ready for something you were never ready to sing; a warmed-up voice still has a ceiling. And it does not need to include your extremes. Touching your highest note to check it still works, first thing, is a habit worth dropping. You are asking the most of the tissue at the moment it is least prepared.

Cooling down, which almost nobody does

After a hard session or a gig, your voice has spent an hour or two in a configuration it never uses for talking. Higher pressure, longer closed phase, more muscular engagement. Left alone it unwinds slowly, and you will be speaking on it the whole time it does.

A cool-down is two or three minutes of the mildest thing you know. Descending hums or slides, quiet, in the low middle of your range, drifting down toward speaking pitch. Nothing high, nothing loud. You are handing the voice back to your speaking mechanism deliberately instead of walking offstage into a shouted conversation in a car park.

It is undramatic. Nothing changes that night. Whatever difference it makes shows up the next morning as an absence — nothing much to report.

Water and sleep, without the folklore

Hydration is a habit rather than a fix. A glass of water in the wings does nothing for the next twenty minutes of singing. Steady drinking across the day is the version that shows up in how singing feels, and the difference is real enough to plan around on a heavy day.

Caffeine and alcohol are worth accounting for on a heavy singing day, and not worth panicking about. Most of the rest of what circulates — the drinks, the gargles, the remedies somebody swears by — is folklore, and none of it is a substitute for the last section of this chapter.

Sleep shows up in a specific way. Short sleep rarely makes a voice sound damaged. It makes it uncoordinated. Onsets get less clean, the top of the range feels further off, the trip through your break gets less reliable. If a session feels inexplicably clumsy, count back through your last two nights before you conclude anything about your technique. Chapter twelve covers the other reasons a session can lie to you.

Load, and the part of the day you are not counting

Most voice fatigue does not come from singing. It comes from talking, and above all from talking over noise. An hour of practice is deliberate and measurable. Four hours in a loud bar is neither, and it asks the same tissue for more at a pressure nobody is watching.

So plan the whole day, not just the practice. Twenty focused minutes three or four times a week, spread out, tends to serve you better than the same total crammed into two long sessions, because the gap between sessions is where recovery happens. If you have a demanding sing coming up, protect your speaking voice the day before. That is a bigger lever than an extra warmup.

Silence is a legitimate part of a practice week. A day off is not falling behind.

Normal tired, and the other thing

Normal fatigue after a good session feels like mild effort. The top few notes are less available. Sustained notes want more attention. Your speaking voice sits a little lower and heavier. It is dull rather than painful, it is the same on both sides, and a night's sleep puts you roughly back where you started. That is a working voice behaving like a working voice.

Then there is the other thing, and it gets said plainly.

Pain is not part of singing. Not a burn, not a sharp catch, not an ache in the throat or one that runs up toward the ear. If singing hurts, stop singing.

Hoarseness that hangs around. Any blood. A voice that disappears suddenly. Any feeling of something obstructing the throat, or of choking. Any change in your voice you cannot explain, and especially one that keeps getting worse. Those are questions for a doctor, and they do not come with a waiting period — sooner is better than later, and you do not need to be sure it is serious before you go.

A doctor who can actually look at the folds is the right person, often an ENT specialist or a laryngologist. A warmup is not. Neither is a herbal drink, an internet remedy, a stricter practice week, or waiting to see. Your singing teacher cannot see inside your throat, and no app can either, including this one. A pitch trace is not a diagnosis. Get looked at, then come back and we will work.

Try this

Open Warmups and run Humming thirds as the first thing in your next practice session. Note the average score on the summary screen. Then run the same exercise again as the last thing, after everything else you had planned. Compare the two numbers.

Holding steady means you paced the session well. A sharp drop means you spent more than you had, and the fix is a shorter session next time, not a harder one. Do this a few times across a fortnight and you will start to recognise the drop-off before the score tells you about it, which is the actual skill.

PART

Reading your own voice

Your range is not one number

Ask a singer what their range is and you get a single answer. Three octaves. E2 to C6. Two and a half, maybe three on a good day. It sounds like a fact about a person, the way height is a fact about a person.

It is not. You have three ranges, and they answer three different questions. Confusing them is behind most of the bad decisions singers make about repertoire — the song learned in the wrong key, the audition piece that falls apart in the second chorus, the note that works alone and never works in context.

Least useful first.

The absolute range

This is the one people quote. Every pitch you can produce, from the lowest sound you can make that still has a pitch to the highest squeak you can find. It is not what the range test measures — the test asks only for notes you would let another person hear, which rules out fry and scrapings — but it is close to what the cited figures on the famous ranges page describe.

It is a real measurement of a real thing. On its own, it is also close to useless.

The bottom edge of your absolute range is usually a fry-edged rumble that a microphone barely registers and a band would bury. The top edge is thin and unsupported — you cannot control it, cannot sustain it, cannot land on it twice in a row. Those notes are yours. They are not fake. But they are display notes rather than working notes, real at the edge in the way the fastest you can possibly sprint is real: available for a few seconds, not how you get to work.

Absolute range is also the most flattering number, which is why it is the one that gets quoted. Nobody says "I have two octaves of usable middle and a fragile fifth on top." They say "three octaves." The number rewards edge-hunting, and edge-hunting is the least likely thing to make you a better singer.

Where it earns its place: it points you in a direction, it gives you something to measure against months from now, and it rules things out. If a melody sits a fourth above anything you can produce at all, you have your answer before you spend three weeks on it.

The comfortable range — your tessitura

Tessitura is the old word for where a voice lives. Not the extremes it can touch. The band of pitches where it sounds like itself with the least work.

You know your tessitura by feel more than by measurement. No test hands it to you: the range test reports a low note, a high note and a voice-type label, and nothing at all about where you live between them. Tessitura is the region where your tone is even and you are not thinking about breath, where you can sing a phrase, breathe, sing another, and another, without the effort piling up on you. It is where you sing when you are humming to yourself and not paying attention. It sits well inside the absolute range, with real distance at both ends.

This is the number that decides what you can sing well. A song does not sit on its highest note. It sits in the band where the melody spends its time, and that band is usually a narrow one — which is why a song with one dramatic high note can be comfortable, and why a song that never leaves your middle can still wreck you if it parks there for four minutes and never lets you off.

Your tessitura also moves in ways your absolute range does not. It shifts with the time of day, with how much you have talked, with how well you slept. Consistent work tends to widen it — not by adding notes at the edges, but because notes that used to cost you stop costing you. That is the change worth watching, and it is invisible if you only track your highest note.

The performable range

Between the two sits the range you can use in front of people, on a particular night, in a particular song.

Narrower than your absolute range, wider than your tessitura, and unlike either of them, contextual. It depends on what came before it in the set. On whether you have to sing loudly. On whether the note is approached by step or by leap. On whether you hold it or pass through it. On whether you also have to look relaxed while doing it.

A test: a note is in your performable range if you would bet on it. Not "I hit it once in practice" — would you put money on it landing, cold, in the third song, with people watching? That question moves the boundary a long way in from the absolute edge, and it moves it honestly.

The top of your performable range sits somewhere below your absolute ceiling, and the bottom sits above your absolute floor. Where exactly is a question only your own betting can answer. Those gaps are not wasted space. They are the margin that makes everything below them sound confident. A singer working at the absolute edge sounds like it, every time.

Holding all three at once

Three ranges, three jobs.

Absolute range tracks change over long stretches and rules out the impossible. Nothing measures it exactly — the range test deliberately stops short of it, counting only notes you would let another person hear. What the test gives you is a repeatable marker near that edge, taken three times across the twelve weeks and left alone in between.

Tessitura chooses songs and decides what key to sing them in. It is the number to know best and the one nobody asks about.

Performable range is for set lists, and for the moment a friend hands you a phone and asks you to sing something. It is the practical answer to "can you do this."

When you look at another singer's cited figure — on the famous ranges page or anywhere else — you are almost always looking at an absolute number, collected across a whole career, in studio conditions, sometimes from recordings made decades apart. It tells you little about where that voice was comfortable, which is the thing you would want to know if you were deciding whether their songs suit you. Later chapters cover how to use that library well. For now, hold the numbers loosely.

The most common self-defeating move here is simple. Someone takes a range test, sees a top note, and starts choosing songs that reach it. The songs fail — not because the note was fake, but because it was never a note to build on. It was the ceiling, and they treated it as the floor of the chorus.

Work from the middle out. The middle is where your voice is honest, where it is safe to build, and where practice compounds. The edges follow the middle. They rarely lead it.

Try this

Take the range test properly warmed up and note the result: a low note, a high note, a voice type. Then open the pitch studio and find two notes by ear, because the test will not find them for you: the highest you could hold for a full phrase without bracing for it, and the lowest that still carries at speaking volume. Read both off the tuner and write them down beside the test result. The gap between the two readings is the whole point — that inner band is your working range, and it is the one to build repertoire on. Save the range test to Progress so the next reading has something to sit next to.

Reading your range test

The range test asks you to sing down until you run out of bottom, then up until you run out of top. It hands back two notes and a label. That is a small amount of information, and it is easy to collect badly. The goal here is not a better number. It is a number that still means something in six weeks, when you take the test again and want to know whether you are comparing two measurements or two moods.

What the test is actually measuring

The test measures the lowest and highest notes you would let another person hear — today, in this room, with the voice you warmed up or did not. That is the entire claim. It is not your absolute range, because the creaks and scrapings at either end are sounds rather than notes. It is not measuring your potential, your talent, or the range you would have on a better day. Those exist. No measurement reaches them.

Which matters, because the two edges are the least stable part of your voice. The middle is boring and reliable; you could sing a comfortable G in your sleep. The bottom and top are where a short night, a dry room, or a slightly different amount of effort moves the result while nothing about you has changed.

Treat the two numbers like a bathroom scale. Real, useful across months, close to meaningless on any single reading.

Taking it honestly

Honest means the same conditions every time, and the same definition of a note every time.

Warm up first, without going long. A cold voice under-reports at both ends, and the bottom in particular needs a few minutes before it speaks cleanly. Run a short set from the warmups library, then test. Do not test after an hour of hard singing either — that reads low for its own reasons. Pick a length and keep using it.

Same distance from the phone. Mic distance changes what the detector hears. Quiet low notes need the mic close or they never register at all, and loud high notes with the mic close can distort into something the pitch detector cannot lock onto. A hand's span from your mouth, held roughly still, is a workable habit. Write down what you chose. You are after repeatability, not perfection.

Count only notes you would let someone else hear. This one rule decides whether the test is worth anything. A note counts if it has an identifiable pitch, arrives without strain, and could be held for a beat or two. It does not count if it is a creak, a squeak, a breathy suggestion of a pitch, or something that arrived along with your neck. Vocal fry is the classic case: most people can make a rattling noise somewhere below their real lowest sung note. It is a legitimate sound. It is not a note you can sing a phrase on, and counting it puts your bottom number somewhere you cannot actually sing.

Stop before the edge, not after it. If a note feels like effort rather than height, you are done. The reading you get by stopping honestly is the one you can reproduce next month; the one you get by grinding out a final semitone is not. And if reaching for the top hurts, that is not a data problem — stop. Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more.

When the result looks wrong

A disappointing result usually has a boring explanation. Work through these before you believe it.

Cold voice. The usual culprit. Retest after a proper warmup before concluding anything.

Time of day. Many voices sit lower and feel thicker in the morning, then loosen over the following hours. Test at roughly the same hour each time, or accept that a morning test and an evening test are two different tests.

Pushing. Effort at the top narrows range rather than extending it. Press, and the mechanism thickens exactly where it needs to thin, so the top note arrives earlier and sounds worse on the way. If your top went down after a stretch of hard practice, over-pressing explains that more often than lost range does.

The room and the phone. A live bathroom, a fan, a fridge, a phone lying face down on a duvet — each of them costs you notes at the quiet end. Test in the same room when you can. Chapter 12 holds the full measurement checklist and goes further into the ways rooms lie to you.

Rushing the edges. Some people sweep through the last few notes too fast for anything to lock on. Give each note near the edge a moment to settle before deciding it is not there.

If two of those are plausible, you do not have a result. Discard it and test another day. A reading you know was compromised should never enter your history as evidence.

The voice-type label

You get a label, and the test calls it the closest fit. That word is doing real work.

The label is a rough classification based on where your two edges fall. As shorthand it earns its keep: a starting key, a direction to browse repertoire, which end of the piano to look at first.

What it is not is an identity, a ceiling, or a verdict. Classical classification uses far more than two numbers — tessitura, timbre, where the transitions sit, how the voice holds up under sustained load — and even then teachers disagree, and singers get reclassified years into a career. A label derived from two extremes on a phone is a first approximation.

The failure mode is letting the label decide what you attempt. A baritone label does not put high notes off-limits. It says your comfortable centre probably sits lower than a tenor's. That centre is the thing that actually decides whether a song fits, and the test does not report it — a low note, a high note and a label are the whole of what comes back. The chapters on choosing songs return to this.

If the label changes between tests, that is usually your edges wobbling across a category boundary rather than your voice changing type. Do not celebrate it, do not mourn it.

Retesting fairly

Fix the protocol first: same warmup, same room, same mic distance, same time of day, same rule about what counts. Then leave it alone. A rough protocol you actually repeat is worth more than a perfect one you manage once.

The programme retests three times: week one for a baseline, the end of week six, and the start of week twelve. Testing more often than that mostly measures noise — and there is a subtler cost, which is that frequent testing quietly becomes daily edge-pushing, and the edges are the worst place to spend practice time. Every range test makes a small demand on the top of your voice. Do not make it a habit.

When you read the range history in Progress, look at the shape of the line across several tests rather than the gap between the last two. Three readings leaning the same way say more than one jump, and one jump usually means a condition changed rather than a voice. Chapter 11 covers how long to wait before concluding anything.

Record the context somewhere too — a line in a notebook is enough. Tested with a cold. Tested after a late night. Tested in a hotel room. In three months you will not remember, and the point on the chart will look like information.

Try this

Open the warmups library, run a short set, then take the range test — this time stopping the moment a note needs effort rather than height, and counting only pitches you could hold for two beats. Compare that reading against your last one in Progress, and note which of the two you took more honestly.

What pitch accuracy actually measures

A number appears at the end of a warmup. Ninety-one percent. Eighty-four. You feel something about it before you understand it, and what you feel is usually wrong, because almost nobody is told what the number is counting.

Here is the mechanic. Many times a second, the app estimates the pitch coming out of your mouth and compares it to the note the exercise asked for. If you are inside a tolerance window around that note — roughly half a semitone either side — that sliver of time counts as a hit. The score is the share of the exercise's scored time you spent inside the window. It is a stopwatch, not a grade: it measures how long you were close enough, not how close.

That distinction changes things immediately. Sit forty cents flat for a whole phrase and you can still score in the nineties, because close enough is a window and not a point. Sing dead centre for most of a phrase and lose the last note entirely, and those seconds outside the window count against you at full weight no matter how good the rest of it sounded.

Cents, and how small they are

The distance between two adjacent keys on a piano — C to C sharp — is one semitone. Split that semitone into a hundred equal slices and each slice is a cent. Cents exist because the ear hears ratios, not hertz. The gap between A2 and A sharp 2 is about seven hertz; the same gap three octaves higher, between A5 and A sharp 5, is about fifty. In hertz those look like different-sized problems. In cents they are the same size: a hundred, both times. That is why the pitch studio reports cents and draws its trace in semitone lanes rather than showing you a raw frequency.

Twenty cents sharp is a fifth of the way to the next key. Written down, that sounds enormous. In motion it is not. On a note that passes in a quarter of a second inside a running line, twenty cents is close to invisible to a listener. Held on a long exposed vowel with a piano underneath, the same twenty cents sits there and curdles. Same measurement, opposite musical consequence.

Which is the first thing the score cannot tell you: where the error was. A tenth of a second outside the window at the start of a phrase and a tenth of a second outside it on the final sustain arrive in the total as the same tenth of a second. Your ears know the difference. Trust them over the number.

Why perfect is neither possible nor wanted

No sung tone holds perfectly still. Breath pressure fluctuates, the folds respond, and the pitch moves in small amounts even on a note that sounds rock solid. But the deeper point is not physical. Dead centre is not where good singing lives.

The tuning system in the app, on your keyboard, and in nearly every backing track you will ever sing over is equal temperament: the octave chopped into twelve identical semitones so that instruments can play in any key. It is a compromise, and it is an audible one. A major third in equal temperament sits wider than the third the overtone series produces on its own. Unaccompanied singers in a good choir tend to drift off equal temperament without ever discussing it, because the chord locks when they do. Their thirds go flat of the piano and the sound gets better. A tuner would mark them down.

Soloists do the same thing for expression. Leading notes get leaned on slightly sharp to increase the pull toward resolution. Blues and soul phrasing lives in the space between the written pitches, and much of what makes those styles sound like themselves would be scored as error.

So the target is not zero. The target is control — being where you meant to be, including when where you meant to be is off centre. An accuracy score measures agreement with equal temperament, and that agreement is a proxy for control rather than the thing itself. It is a good proxy while you are learning. It gets less useful as you get better.

Three different ways to be off

Lumping every deviation under "out of tune" is why the number frustrates people. There are distinct failures with distinct causes, and the shape of the trace tells you which one you have.

Drift is slow and directional. You start the phrase where you intended and arrive somewhere else — usually flat, usually late, sliding rather than stepping. On the trace it looks like a line leaning away from the target over several seconds. Drift is a breath and energy story, not a hearing story: pressure falls off as the air runs out and the note sags with it. If your first four bars are clean and your last two slide south, the problem is not in your ears.

Scooping is fast and local. You arrive underneath the note and slide up into it, a swoop at the onset half a beat long or less, then a stable centred pitch for the rest. The trace shows a hook at the front of each note. Scooping is a note-starting habit. Sometimes it is deliberate style; more often it is hedging, the voice searching for the pitch instead of landing on it. Deliberate scoops are a real technique. Involuntary ones cost you scored time on every onset, which is why a singer who sounds fine to a friend in the room can score lower than the room suggests. The fix lives in onset work, not ear training.

Genuinely out is stable and wrong. You hold a steady, confident pitch, and it sits a distance from the target and stays there. That is a hearing or reference problem: you are matching something other than the note asked for — a harmonic, a neighbouring scale degree, a pitch memory from another key. Being consistently out by a whole semitone usually means you grabbed the wrong note, not that your tuning is poor.

Drift wants breath. Scooping wants onsets. Genuinely out wants ear work. The trace tells you which of the three to spend tomorrow's twenty minutes on, and that is worth more than the percentage attached to it.

How vibrato reads to a tuner

Vibrato is a pitch oscillation: the note rises and falls a few times a second around a centre. Freeze it at any instant and the measured pitch is somewhere out on that swing, often nowhere near the target.

A tuner has no concept of a centre averaged over time. It reports where you are right now. So an even vibrato registers as a continuous alternation of sharp and flat, and a wide one spends part of every cycle outside the tolerance window — scored time lost without a note being sung wrong. The ear hears one pitch with warmth on it. The meter hears a note that keeps leaving.

On the trace this is unmistakable once you have seen it: a regular, even wave centred on the target line. That is vibrato, and it is fine. An irregular wander with no steady period and no consistent centre is not vibrato — that is an unsteady note, a different problem with a different fix. The tell is regularity, in both rate and width.

So if you have an established vibrato, do not chase a higher percentage on sustained notes. Judge sustains by the shape of the wave, and take the percentage seriously on scales, arpeggios and pitch matching, where vibrato is not doing the work.

None of this makes the number useless. It makes it a reading rather than a grade — one instrument with known blind spots, best used to point you at which of the three failures you actually have.

Try this

Open the pitch studio and pick a note on the keyboard in Target practice. Sing it eight times: four with your normal onset, four landing on the note from just above rather than sliding up from underneath. Watch two things — the cents readout at the instant the sound starts, and the front edge of each note on the trace. If the hook at the front disappears when you come from above, scooping is most of what your score has been counting.

Finding your own break on the chart

Charts put a tenor's first transition around C4 or D4 and a second around F#4 or G4, and a soprano's around F#4 or G4 and then C#5 or D5. Those figures are a rough middle of a wide spread, and a rough middle is useful right up until you try to sing one. Your transition sits where your folds and your vocal tract put it. It can land either side of the printed number, it can be a zone rather than a single note, and it can shift with the vowel, the volume, and how long you have been awake.

So the job is not to look up your break. It is to find it, write down a note name, and keep checking whether that name still holds.

The slow slide, and what to watch

The pitch studio is the room for this. A slow ascending slide on one sound, quiet, with the trace running so you can watch and listen at the same time.

Start in your middle, somewhere you can talk-sing without thinking. Use an "ng" as in "sung", or a soft "oo". Both keep the sound small, which is the point. At low volume the transition shows up as a change in quality rather than a crash, so you can observe it instead of fighting it. Slide up slowly enough that a whole step takes four seconds. Then slide back down.

Do that three or four times, watching three separate things.

The first is what you hear. Somewhere on the way up the sound stops being one continuous colour. It may thin, hollow out, brighten suddenly, or flip outright to something lighter and less connected. That change has an address. Slide again, slower, and name the note it happens on.

The second is what you feel. Below the transition the buzz sits low and broad — chest, throat, the front of the face. Above it, it climbs and narrows. Somewhere in between, the effort jumps: the same quiet slide suddenly asks more of you, or you catch yourself bracing. For many singers that jump arrives a semitone or two below the audible change, which makes it the earlier of the two warnings.

The third is the trace. Watch the pitch line through that zone. A slide that is clean below and clean above will often go scruffy in the middle — a wobble, a small step, a flat patch where the voice hesitates before committing. The trace will not tell you why. It is good at telling you where, and it does not flatter you the way memory does.

Three or four passes usually give you a zone rather than a note: "it gets strange between D4 and F4." That zone is the honest answer. Write down its lowest note, because that is where preparation has to start, not where the trouble becomes obvious.

Then do it on more than one vowel. Try "ee", then "ah". If your transition sits higher on the closed vowel than the open one, that is not sloppy measurement. It is a real property of your voice, and knowing it is worth more than one tidy number.

What the range test adds

The range test is doing a different job. It is hunting for the outer edges, but it leaves fingerprints on the way through.

Take it the way chapter 8 describes, then look back at where the ascent got difficult. The note where you had to change something to keep going often sits at or just above your transition. And if your range history in Progress shows a top note parked in the same place across retests while the climb up to it has got easier, that flat line may be the transition rather than a true ceiling.

If you have Coach, the per-note accuracy breakdown is the sharpest tool here. A band three or four semitones wide where notes land in tune less often than their neighbours is worth a second look: your ear knows the pitch, and the coordination is what is wobbling. Treat a dip in the middle of an otherwise even profile as a transition until something else explains it better.

Belt ceilings, in practice

People talk about a belt ceiling as if it were a lid. It is closer to a point past which one particular way of singing stops being available.

Belting carries chest-dominant weight above where that weight wants to go, held together by how you shape the tract. Every voice has a pitch where that coordination comes apart — the sound gets shouty, the pitch goes sharp or unsteady, and one more semitone costs far more than the last one did. That pitch is your belt ceiling for today, at that volume, on that vowel.

Two things matter about it. First, it is not the top of your range. Head voice goes higher, often much higher, and the range test measures the higher figure. If your test says A4 and the belt gives out at E4, both numbers are true and they are describing different things.

Second, a belt ceiling is a coordination limit rather than a wall to push through. Hammering the note just above it tends to end a session worse than it started. Chapter 17 covers adding height properly. For now, find where the cost curve turns steep and treat it as information: the top of the range you can write into a song and expect to sing four nights running.

Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more.

Head voice or falsetto

Above the transition you have more than one option, and telling them apart matters more than the labels suggest.

By ear, falsetto is the airier of the two. Breath sits in the tone, it does not carry far, and its dynamic range is narrow — push for volume and it breaks or flips down. It tends to sound detached from the notes below, as though a different instrument took over. Head voice is cleaner and more compact. It has a core. It can crescendo without falling apart, and it connects downward: you can slide from head voice into your middle without a seam.

By feel, falsetto is easy in a hollow way. Very little resistance, very little sensation below the throat, as though the sound is happening without you. Head voice at the same pitch feels gathered. The breath meets something, and the tone is held rather than released.

The recorder settles arguments the mirror cannot. Sing the same three or four notes above your transition twice — once deliberately breathy and light, once keeping the tone compact and connected — then A/B the two takes. A difference that disappears while you are singing is often obvious on playback.

One caution: falsetto is not a failure state. It is a real mechanism with real uses, and plenty of singing you admire lives there. The reason to tell them apart is that they answer to different practice. Working on connection while you are in falsetto is tuning the wrong string.

Write down three notes — where the effort jumps, where the sound changes, and where the belt stops being available. Check them again at the range retest at the end of week six. They will not match exactly, because voices are noisier than that. The direction they drift tells you more than any single reading.

Try this

In the pitch studio, sing a slow "ng" slide upward at low volume, three times, and note the pitch where the trace first gets scruffy — not where the sound flips, but where it starts to hesitate. That is where your preparation needs to begin.

Tracking change over months, not days

Your voice on Tuesday is not the instrument it was on Monday. You slept differently, drank a different amount of water, spoke for four hours instead of one, sang before breakfast instead of after dinner. None of that says anything about your singing. All of it shows up in your numbers.

This is the most common way singers misread themselves. A range test comes back lower on top than last week, and you spend the evening deciding you have gone backwards. Or it comes back higher and you decide the work is paying off. Both conclusions rest on the same amount of evidence, which is none.

What noise looks like

Noise is variation with no cause you can act on. In a voice it arrives from everywhere: hydration, sleep, how recently you talked, allergies, time of day, how warm you got before testing, how close the phone sat, how strictly you held to counting only notes you would let another person hear. Any one of those can move where the test lands. Stacked on a bad day, they move it further.

The consequence is simple. A single session tells you about that session. It tells you very little about your voice as an instrument. To learn about the instrument you need enough sessions that the noise cancels itself out and whatever is genuinely moving stays visible underneath.

There is no clean rule for how many that is. A workable habit: draw no conclusion from fewer than five or six scored sessions spread across at least a month, and treat any reading that sits far outside the others as suspect until it repeats. One outstanding day is not a new ceiling. One terrible day is not a decline.

Two things move at different speeds

Separate what you are watching, because different parts of your singing run on different clocks.

Coordination moves fastest. Onsets getting cleaner, a scale holding in tune where it used to sag, the transition through your break costing you less panic — that is motor learning, and repetition is what feeds it. Your pitch studio accuracy and your warmup scores are mostly measuring coordination, which is why they tend to shift first.

Range boundaries move slowest, and sometimes not at all. The extreme top and bottom of what you can produce are shaped by your anatomy, and practice does not redesign anatomy. What practice can change is your access to notes that were already available but badly coordinated, and the comfort with which you sit in the middle of what you have. Someone who describes gaining a fifth on top over a year has usually gained control, not new notes.

So if you want evidence that something is working after three weeks, look at accuracy and consistency rather than at the edges of your range test. A flat range history alongside tighter per-note accuracy across your middle octave is the better result of the two, even if it makes a duller screenshot.

What the progress charts can tell you

Progress keeps your range history under Your voice, and your warmup and studio sessions accumulate scores over time. In Pro, Coach tracks accuracy note by note, so you can see which specific pitches keep letting you down.

That data is good for four things.

Trend direction over a long window. Six months of range tests, taken with reasonable consistency, will show whether your low note and your high note have moved or held. That is a real finding. It is the only thing the test reports, alongside a voice-type label, and it is not a reading of where you are comfortable.

Which notes are persistently weak. A note that reads badly once was a bad rep. A note that reads badly across twenty sessions is a coordination gap, and it is exactly what Coach's daily plan keeps circling back to.

Whether you are actually practising. Streaks and session counts are the boring numbers on the page and also the hardest to argue with. Intending to practise remembers itself as practising. The practice calendar does not.

Before-and-after on a recording. A number gives you one dimension. A take from twelve weeks ago, lined up against one from today with A/B compare in the Recorder, gives you tone, steadiness, phrasing and confidence at once. It is the least precise measurement in the app and often the most honest.

What the charts cannot tell you

They cannot tell you why a number moved. The app measures what came out of you in a room, through a microphone, on a particular morning. It has no idea you were tired.

They cannot compare you to anyone else. Your range history is a record of you against you. Someone else's numbers came from a different room and a different idea of where a usable note stops.

They cannot separate a good session from a lucky one. Voices have good days, and a day where everything lands is worth enjoying. If it does not repeat, treat it as the top end of your normal variation rather than a new baseline.

And they cannot tell you when something other than practice is behind a change. Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more. A chart cannot answer that question and neither can more practice.

How to actually watch a trend

A few habits make the data worth keeping.

Keep the conditions comparable — same rough time of day, same room, same distance from the phone, same amount of warmup — because you are not chasing the highest number, you are after one that can sit beside the last. Chapter twelve has the full checklist and the ways a reading goes wrong.

Test on the schedule, not on a feeling. Retesting because your voice feels great today guarantees a history of your best days rather than your typical ones. The program asks for three range tests — week one, the end of week six, and the start of week twelve — and three honest readings beat a dozen opportunistic ones.

Write down the context. One line — slept badly, second day of a cold, sang for two hours last night — turns an unexplained dip into an explained one when you look back in March.

Read the last three months, not the last three sessions. When you open your range history, resist the rightmost point. Cover it with your thumb if you have to. Ask instead whether the middle of this cloud of points sits higher, lower, or level with where it was two months ago.

Give it time before you change your practice on the strength of what you see. Four weeks on the same thing with no movement in the relevant numbers is worth examining. Six days is weather, not climate.

The uncomfortable part

Long stretches of a trained voice look like nothing is happening. You practise four times a week for two months and the chart is a horizontal band with scatter in it. That is ordinary, and it is not evidence the practice is wasted. Coordination consolidates quietly, and the change often shows up first in things the app never measures: less dread before the high note, a phrase you get through without thinking about air, a song that used to feel like a fight.

The chart is a slow instrument pointed at a slow process. Read over months, it is worth trusting. Checked daily, it mostly reports what you had for dinner.

Try this

Open Progress, find your range history under Your voice, and read only the points older than four weeks — cover the recent end. Ask whether that older cluster sits higher, lower, or level with the one before it. Write down what you see, so next month you have something to compare it against.

When the numbers lie to you

Every number the app shows you measures two things at once: your singing, and the conditions you sang in. Usually the second is quiet enough to ignore. Sometimes it is louder than the first, and you end up reading the room instead of your voice.

Bad data does not feel like bad data. It feels like a verdict. A range test that comes back a third short of last month's does not announce itself as a microphone problem — it announces itself as loss. And a singer who believes they have lost a third starts pushing to get it back, which is the wrong response to a measurement error.

What the app is actually hearing

The pitch trace, the range test scoring, the per-note accuracy Pro reports — all of it runs on your device, from your device's microphone. Nothing is uploaded, and no engineer is cleaning up the signal on the way. That is good for privacy and good for speed. It also means the input is exactly as good as your room and your setup make it.

Pitch detection works by finding a repeating pattern in the incoming sound and reporting how often it repeats. That is reliable when the loudest thing in the signal is one voice on one note. It gets less reliable as other steady sounds compete, as the signal drops toward the noise floor, and at the edges of your range where your tone carries less energy in the fundamental.

Which gives you a short list of things that can corrupt a session.

Room noise

A fan. An air conditioner. Traffic through a single-glazed window, a fridge compressor, a laptop under load, someone watching television in the next room. Low steady noise is the worst kind, because it sits in the same neighbourhood as your low notes and never stops long enough for the app to find a gap.

The symptom: the trace shows a pitch when you are not singing, or your low notes read as jittery while your middle sounds fine. If your bottom three or four notes look wild and the rest of your range looks normal, suspect the room before you suspect your throat.

Test it in ten seconds. Open the pitch studio, stay silent, and watch. A clean setup shows nothing. If the trace finds notes in your silence, it will find them while you sing too, and mix them into the reading.

Mic placement and distance

Phone microphones are close-range devices with automatic gain. Move from a hand's width to arm's length and the app is hearing a different instrument.

Too close and loud notes clip: the waveform flattens, the pitch estimate wobbles, and your best-supported belt reads as the least accurate note of the session. Too far and quiet notes fall under the noise floor, so the top of your head voice and the bottom of your chest voice never register at all. That is the classic false range test.

You sang the note, the app did not hear it, and the result says you cannot reach it.

Most people move the phone as they sing without noticing. You lift it as you go up. You bring it in on the quiet notes. Automatic gain rides those movements, and your loudness data becomes a record of your arm.

Pick a distance — roughly a foot, a hand's span and a bit — and a position, and keep both the same every time. Propped on a stand or a stack of books beats held in a hand, because a stack of books does not flinch on the high note. Angle it slightly off-axis so your breath is not blasting into the capsule, which keeps plosives and breath noise out of the signal.

Headphone bleed

This one produces the strangest readings. If you practise with open-back headphones, or earbuds turned up, or a backing track playing out loud, the microphone hears the track as well as you.

What comes back is neither you nor the track but a blend of the two. Sustained notes read as accurate while you are silent. Your scores drift toward whatever the reference tone is doing, which flatters you uselessly. In melody echo and pitch-match, the exercise's own reference can leak back in and score itself.

Closed-back headphones or in-ears at a moderate volume fix it. If you are on speakers, the numbers from that session are decoration. Enjoy the practice and do not log the result.

Time of day, and the body you woke up in

Your voice is not the same instrument at 7am as at 2pm. Plenty of singers find the bottom is available first thing while the top takes hours to arrive; others run the other way. Either pattern is worth learning about yourself. Neither is worth generalising from one morning.

Add the ordinary variables. How much you slept. How much you talked yesterday. Whether you were shouting over a band on Friday, where you are in a menstrual cycle, how much water you have had, how dry the air is. None of these is a fault. All of them move the numbers.

The consequence is simple. A range test taken cold at 7am and compared against one taken warm at 3pm is not a comparison. It is two different measurements wearing the same label. If you are tracking change across months, test at roughly the same time of day, after roughly the same warmup, and note anything unusual about the day.

Colds, allergies, and the weeks around them

Congestion changes what you hear more than it changes what comes out. A blocked nose alters the resonance you perceive through your own skull, so you can feel badly out of tune while the trace says you are fine, or feel fine while you drift.

The practical part is easy: put the numbers aside. A range test taken with a cold measures the cold. Go by how you feel about singing at all, and start measuring again when you are back to ordinary.

Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more.

Spotting a bad session before you act on it

You do not need to analyse every practice. You need one question before you let a number change your plan: *was this session measurable?*

A pre-flight, thirty seconds at most:

- Silence check in the pitch studio. Does the trace stay empty when you do?
- Same distance, same position, phone propped rather than held.
- Headphones in, closed or in-ear, volume moderate.
- Warm up before you test, not after.
- Note the time and how you feel — rough, tired, congested, fine.

And a few tells that a session went wrong mid-flight. Accuracy uniformly bad across your whole range rather than worse in the usual places, which points at the setup more often than the voice, because voices do not degrade evenly. Low notes erratic, middle clean. Notes you sang last week that do not appear at all. A trace that lags or stutters. Everything reading sharp or flat by the same amount on every note, which usually means you are singing against a reference you cannot hear properly.

When one of those shows up, stop treating the session as data. Fix the setup, or accept that today was practice rather than measurement, and do not save the range test. One discarded test costs you nothing. One believed bad test sends you chasing a problem that was never in your voice.

Try this

Before your next range test, open the pitch studio and stay silent for fifteen seconds while you watch the trace. If it reports any pitch at all in your silence, find the noise and kill it before you test. Otherwise your low notes are measuring your room.

PART

The twelve-week program

How the twelve weeks work

The next seven chapters are a plan you can actually run: twelve weeks, four blocks of three weeks each, three or four sessions a week, twenty minutes a session.

That is roughly fourteen hours of singing across three months. It is deliberately not very much, because the thing that decides whether this works is not how hard any single session is. It is whether you are still doing it in week nine.

Why blocks

Each block has one job, and the blocks are ordered so that each one rests on the last.

Block one, weeks 1–3: settle. Baseline measurements, breath work, gentle warmups, and time spent on the part of your voice that already behaves. No target beyond honest data and a habit.

Block two, weeks 4–6: negotiate. Work the middle until it stops surprising you, then start moving through the transition — sirens and slides, quietly, with vowel doing the work instead of effort. The block closes with a range retest at the end of week six.

Block three, weeks 7–9: extend. Height without weight at the top, then the bottom, which asks for release rather than push. This block carries the strictest limits on volume and repetition.

Block four, weeks 10–12: integrate. Long phrases, then real songs, then a retest at the start of week twelve against what you recorded in week one.

The chapters that follow move in fortnights rather than threes, because a fortnight is the unit you actually plan around and because the work overlaps the block seams. Nothing breaks if your week six looks a bit like your week seven. The blocks are the shape; the weeks are how you live in it.

The order is not the one most people would choose. Left alone, singers go straight for the top, because the top is the part they can hear missing. Starting in the middle and only reaching the edges in weeks 7 through 9 is the whole point of the sequence. The middle is where coordination gets built cheaply. The edges are where it either shows up or does not.

What twenty minutes looks like

A session that short needs a shape, or it disappears into fiddling with settings.

Four or five minutes of breath work. Six to eight minutes of warmups from whichever pack the week calls for. Five or so minutes on the specific thing that week is about — pitch-match, sirens, sustain, a phrase from a song. Then a minute or two quieter and lower than everything before it.

That last minute matters more than its length suggests. Ending on something easy leaves the voice in a better place than ending on the hardest thing you did, and it means your last impression of the session is a good sound rather than a struggle.

If you only have ten minutes, do breath and warmups and stop. A short honest session counts. A long distracted one does not.

Three or four sessions a week

Not seven. Voices adapt in the gaps as much as in the work, and daily singing at this stage tends to mean either sessions too short to do anything or fatigue that accumulates faster than you notice it.

Three is the floor. Four is better. Five is fine if the sessions stay at twenty minutes and stay gentle. Spread them out — Monday, Wednesday, Friday beats three sessions crammed into a weekend, because the point is repeated exposure, not total volume.

Coach, on the Progress page, reads your recent scored sessions and range tests and builds a short plan around whichever area came back weakest: ear training, warmups, or song work. It is arithmetic on your own numbers, not a second opinion. Free accounts see the first step of the plan; Pro shows all of it. Either way, it is a reasonable way to fill the middle of a session without deciding for yourself every time. It does not replace the block structure — Coach picks what to drill, the program decides what the fortnight is for.

Before you start: what to record

Do this in one sitting, before week one, and do not skip it because you are impatient to sing. Everything in week eleven depends on having something honest to compare against, and you cannot go back and collect it later.

A range test, taken by the rules. Warmed up, same room, counting only notes you would let someone else hear. Note the conditions somewhere — time of day, how you slept, whether you had a cold.

A recorded take. One verse and one chorus of a song you already know well, in a key that is comfortable, sung the way you would actually sing it. Not your hardest song. Not a performance. One honest pass, saved in the Recorder so you can find it in twelve weeks. It is the most useful thing on this list and the easiest to talk yourself out of.

A sustain test. From Breath. One number, thirty seconds of your attention.

A sentence about why. What you want to be able to do that you currently cannot — sing the second chorus without running out, get through a bridge without the break showing, sing at all in front of someone. Write it down. Wants drift over three months, and in week eleven that sentence will tell you whether you got closer to the thing you actually cared about better than any chart can.

That is the baseline. Twenty minutes of setup for three months of comparison.

Skipping, repeating, going back

The plan will not survive contact with your life. Here is what to do when it does not.

Missed one session. Nothing. Carry on where you were. Do not double up to catch it.

Missed a week. Repeat the week you missed rather than jumping to where the calendar says you should be. You lose seven days and keep the sequence intact, which is the better trade.

Missed two weeks or more. Go back to the start of the block you were in. If you were in week eight, restart at week seven. What fades first over a break is coordination rather than strength, and coordination is exactly what the earlier week rebuilds.

A week that is not working. Repeat it. This is normal and it is not failure. Weeks 5 and 6 are the ones people repeat most, because the transition takes as long as it takes and no amount of pushing shortens it. Repeating a fortnight costs far less than carrying an unbuilt skill into the block that assumes it.

A session that has got harder for no reason you can name. Effort where there was none, a voice that is tired after ten minutes when it used to last twenty. Usually that is load — sleep, a long week of talking, more singing than you logged. Go back a block, work quieter, and give it a fortnight.

Pain, blood, or a voice that will not clear. This is a different category and the program has nothing to offer it. Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more.

Ahead of schedule. Do not skip forward. A week that feels easy is a week you get to do well rather than a week to be got past.

Two rules override all of the above. First, when in doubt, go back rather than forward — nothing in twelve weeks is lost by spending an extra fortnight in the middle of your range. Second, do not retest more than the plan asks. There are exactly three range tests in twelve weeks: the baseline in week one, one at the end of week six, and one at the start of week twelve. Do not retest between those points. Testing weekly turns the top of your voice into a practice room, and chapter 11 explains why the readings would not tell you much anyway.

Try this

Before your first session, take a range test, then record one verse and chorus of a song you already know into the Recorder — and leave both alone. In week eleven you will record that same song again and put the two takes side by side in A/B compare. Whatever sits between them is the only honest evidence you will have.

Weeks 1 and 2: baseline and breath

The first two weeks are not about singing better. They are about two things everything after them depends on: a set of measurements you can trust, and the habit of turning up.

That is a frustrating thing to read on day one. You want to work. But a first reading taken in the middle of an effort is not repeatable, and eleven weeks later you will have nothing solid to compare against — the number will have been a good day or a stubborn one, and you will not remember which. A careful baseline gives you a fixed point. The breath habit gives you something small enough that you might still be doing it in week nine.

So the instruction for these two weeks is: do less than you can. Everything here is under-loaded on purpose.

Day one: the baseline session

Set aside about thirty minutes. It is the longest session in the block, and you only do it once.

Warm up first, gently. Five to ten minutes from one of the shorter warmup packs. Do not skip this to get a "true" cold reading. A cold voice under-reports at both ends, and an under-report is not honesty — it is a different kind of wrong number.

Take the range test. Take it the way chapter 8 describes: the room you expect to practise in, a consistent distance from the phone, counting only notes you would let another person hear. Stop the moment a note becomes effort rather than height. This reading anchors the whole twelve weeks, and an inflated anchor is worse than a modest one. It makes every honest reading afterwards look like a decline.

Record the context, not just the number. Time of day. Room. How much sleep. Whether you were getting over something. Whether you had been talking for eight hours. In eleven weeks you will look at this point on the chart in Progress and remember none of it, and a number without its conditions reads as fact.

Record a take. This is the part people skip, and the part that will matter most in week 12. Pick a short piece of something you already know by heart — thirty to sixty seconds, a verse and a chorus, in a key that sits comfortably. Sing it once, at normal volume, without warming up further and without trying to make it good. Save it in the Recorder and rename it so you can find it in three months.

Two rules about that take. Do not do six passes and keep the best one. A best-of-six baseline held up against a single week-12 take is a rigged comparison, and you are the only person it fools. And do not listen back critically today. The first reaction to your own recorded voice is mostly unfamiliarity, and on day one there is nothing in it you can use. File it and move on.

Pick one song you would like to be able to sing by the end, and write it down. You are not going to work on it. It is there so the exercises have a destination.

The shape of the next two weeks

After day one, sessions run twenty minutes, three or four times a week — the same rhythm as the rest of the program. A session looks like this:

- **Breath work, five minutes.** Every session.
- **A gentle warmup pack, ten minutes.** Sung along with, scored, in the middle of your range.
- **Five minutes in Ear training** — pitch match, or the higher-or-lower game — or a second easy warmup if you would rather sing.

That is the whole prescription. It should feel too easy. If you finish a session feeling like you have been to the gym, you have done this block wrong.

The other piece is breath work on the days you do not sing. Five minutes, no warm voice required, and the one thing here worth doing seven days a week.

The breath fortnight

Breath is the first block's real subject because it changes slowly and it barely cares how your voice feels that day. You can do useful breath work at eleven at night, in a room you would never sing in.

Use the three drills in Breath, in a fixed order.

Box breathing first, two or three minutes. In for four, hold four, out for four, hold four. This is not singing practice. It is the part that teaches you to stop gasping, and to tell the difference between having no air and only feeling like you have none. Singers tend to be working with more air than they think.

The Farinelli drill next. Inhale over a slow count, hold, exhale over a slow count, and extend the counts across the fortnight. What you want is an unhurried exhale that does not collapse at the end. Watch for one specific failure: a smooth first half, then a rush through the last two counts. That rush is the same thing that shows up in singing as a phrase running out before the line does. When it happens, shorten the count instead of muscling through it.

The sustain test last, once or twice, not more. Treat it as a measurement rather than an event. You are not setting a personal record, and pushing past the point where you feel lightheaded buys you nothing. If you feel dizzy at all, stop the drill and breathe normally.

Across two weeks, watch the shape of the exhale rather than the number on the sustain test. Does it stay even for longer? Have you stopped hauling in a huge, shoulder-lifting breath before every phrase? Those are the things worth tracking while the numbers are still flat.

Gentle warmups only, and what "gentle" means

Stay in the middle of your range. Use the easier packs — lip trills, hums, five-note patterns, slides that do not cross your break. Sing at conversational volume. If a pattern climbs into a register change and starts to feel like work, stop that repetition and restart the pack lower, or switch packs.

Three things to leave alone in this block:

Do not chase the top. Weeks 7 and 8 are for that, with limits attached, and its position in the program is not an accident. The top of the range is the easiest place to substitute force for technique, and an untrained fortnight is the worst time to find that out.

Do not belt. Volume turns an easy session into a hard one and gives you nothing you can measure.

Do not retest your range. Once, on day one, and then not again until the program schedules it — chapter 13 explains why. A retest taken on impulse contaminates the baseline you just spent thirty minutes taking honestly.

What day fourteen looks like

Not a bigger range. Not a better take. The actual list:

- A baseline range reading you know you took honestly, with the conditions written down.
- One recorded take you did not curate.
- Breath work on most days, and enough repetitions of the Farinelli drill that you know what your exhale does at the end of a count.
- Six to eight singing sessions behind you, none of which left you hoarse. Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more.
- A streak in Progress that is intact, or close.

If a session gets missed, miss it. Do not stack two to make up the difference. That turns a rest day into a heavy day, which is the one thing this block is built to avoid. Pick it up next time.

And if day fourteen arrives and it feels as though nothing has happened, that is the expected result rather than a failure of the plan. The middle-voice work starts in week 3, and it starts from a fixed point instead of a guess.

Try this

Today: warm up for five minutes, take the range test once, and record a single unedited verse and chorus in the Recorder. Rename it "baseline" and do not listen back. Then check one thing — that you stopped the range test at the last note that felt like height rather than effort. That is the number the next eleven weeks get compared against.

Weeks 3 and 4: the middle voice

Weeks 1 and 2 bought you a baseline and the start of a habit. Neither was singing, exactly. Now you sing — and the surprise is that you spend the next two weeks in the least impressive part of your voice.

Not the note at the top you almost have. Not the low one that rumbles when the room is quiet. The handful of notes you talk in, the ones you hum without deciding to.

Start where the voice already works

The edges of your range hold all the interesting questions, which is exactly why they are the wrong place to answer them. Up top and down low, coordination is fragile and small errors get large. A fraction too much air, a grabbed onset, a vowel a few degrees off — the note cracks, or it arrives with a wobble you have to muscle through. You cannot tell which mistake did it, because they all happened at once.

The middle has none of that noise. Closure comes easily there, the air demand is modest, and you are clear of the transition in both directions. Whatever you do in the middle, you can hear, because nothing else is shouting over it. That makes it the one place in your range where you can practise a *quality* instead of surviving a *note*.

There is a blunter reason too. Songs live in the middle. Verses sit there, most of any chorus sits there, and the high note everyone remembers is two seconds of a three-minute song. A middle that behaves carries the whole performance.

So find your working notes. Do not go looking for them in your range test — that gives you a low note, a high note and a voice type, and none of the three is the block you want. Use the practical version instead: the notes you can sing at speaking volume, right now, cold, with no preparation and nothing to warm up first. Keep the middle five or six of those. Almost nothing you practise over the next two weeks should leave that block.

Steady tone

Steady means three things at once, and they fail separately. Pitch that does not drift across the note. Volume that does not sag as the breath runs down. Colour that does not change halfway through, so the note ends sounding like the note that started. The third is the hardest, and it is the one most people never think to listen for.

Work at conversational volume, or a little above. Loud is not the goal here, and loud hides problems — a big sound covers small instabilities the way a noisy room covers a hum.

The test that matters is not whether you can produce one good note. It is whether you can produce the same one twice. Sing a five-note pattern from a warmup pack three times in a row and compare the pitch traces. If the three passes look like each other, you have something. If the first is clean and the next two wander, you got lucky once. Repeatability is the actual skill.

Clean onsets

An onset is how a note starts — the first instant of it, the part nobody practises. It goes one of three ways, and the ways are physical, not stylistic.

Air can arrive before the folds close, and the note begins with a breathy *h*, a small fog ahead of the pitch. The folds can close first, so pressure builds behind a shut door and the note begins with a click. Or the two arrive together and the note is simply *there*: at pitch, at full value, no swoop up from underneath.

The third is what you are training. Not because the others are forbidden — a breathy onset is a genuine expressive choice, and a firm one has its uses — but because you want the balanced onset as your default, the thing your body does when you are not thinking about it.

Week 3 is the right time for it, because onsets are the most repeated event in singing. Every phrase begins with one. Every note in a detached pattern is one. If your default is to hedge, to find the pitch by sliding up to it rather than landing on it, that costs you on every note you sing — in a way no single number in your results will ever show you.

Practise it plainly. One note in your middle, sung five times: short, clean, a full stop and a fresh breath between each. Then run the recorder and listen back. Onsets are far more audible in playback than in your own head, where you hear yourself partly through your own skull.

Sustain

Long notes, one vowel, moderate length. This is not the breath sustain test, which is a measurement. This is a musical exercise, and the goal is *even*, not *long*.

Pick a modest target. Eight seconds that hold their pitch, volume and colour are worth more than fourteen that go thin and flat over the last four. When a sustain sags at the end, the cause is usually pressure falling away rather than anything happening at the folds — which is why the breath work from weeks 1 and 2 does not stop now. It just gets a note attached to it.

Pitch-match and interval work

Ear work belongs in this block for the same reason everything else does: in the middle, production is not fighting you, so what you hear is a clean signal about your hearing rather than a report on your effort.

Three or four minutes a session, every session. These are short exercises, and the point is meeting them often rather than sitting down with them once a week.

In pitch-match, watch something the tuner does not report on its own: how long it takes you to lock on. Landing on the note in a quarter of a second and hunting for a second and a half both end up counted as a match. They are not the same skill, and the second one is the one that turns into scooping when you sing.

The interval game and higher-or-lower ask you to recognise, not to sing, which makes them the part of the program you can do on a train or in a queue. Melody echo is the bridge — hear it, hold it, produce it.

Keep the echoes inside your middle, so the exercise tests your memory instead of your range.

The shape of the two weeks

Three or four sessions a week, twenty minutes each. A workable split: a few minutes of breath, eight to ten minutes of warmups sitting in your middle, three or four minutes of ear training, and the last stretch on a recorded sustain or a short phrase.

Two rules for this block in particular.

Do not retest your range — the next one is scheduled for the end of week 6, and a check at the end of week 4 would mostly tell you what kind of morning you had.

And repeat the same small set of exercises rather than sampling new ones. Novelty feels like progress and rarely is. You want that five-note pattern to get boring. Boring is what reliable feels like from the inside.

Some days the middle will feel thick, or the onsets will keep clicking, or the tone will not settle. Drop the volume, shorten the session, do the same material at half power — coordination still trains quietly. What you should not do is push harder into a voice that is not answering.

Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more.

The aim of these two weeks is not a bigger voice. It is a middle that does the same thing twice.

Try this

In Warmups, pick one five-note pattern that sits in your comfortable middle and sing it three times in a row at conversational volume. Watch whether the three pitch traces lie on top of each other — not whether any single one of them was good.

Weeks 5 and 6: through the break

For four weeks you have worked where your voice already agrees with you. Weeks 5 and 6 move to the place where it does not.

The instinct at a break is to do more — more air, more volume, more determination — and each of those makes the crossing harder. So the work here is smaller and quieter than what came before. If it feels like you have been handed an easier block, you have read it correctly. Easy is the method, not a concession.

You know roughly where your transition sits from chapter 10. Take the lower edge of that zone — the note where the effort first jumps, not where the sound flips — and write it somewhere you will see it. Every exercise in these two weeks aims at the four or five semitones either side of it.

The shape of a session

Three or four sessions a week, twenty minutes, same as the earlier blocks. The internal split changes.

Give the first five minutes to breath and gentle onsets: box breathing from Breath, then two or three minutes of the lightest warmup pack you have, staying entirely below your transition. You are not warming up to the break. You are warming up to the approach.

The middle ten minutes are the work itself — sirens and slides through the zone, then vowel work on sustained notes inside it. Two exercise types, done attentively.

The last five minutes go back down. Sing something comfortable in your middle: a warmup you know well, or a phrase from a song in Songs transposed into your easy range. Ending a session at the top of your effort is how you carry tension into tomorrow.

Twice across the two weeks, record a single siren in the Recorder. Not every session. You want two or three reference takes, not a pile of them.

Sirens and slides

A siren is a slow glide up and back down on one continuous sound, with no break in the tone. It is the most useful exercise in this book, and it is usually done too fast and too loud.

Start on "ng", the sound chapter 10 gave you — tongue up, mouth relaxed, sound small. It is hard to push on an "ng", which is what makes it the easiest place to begin. Start two or three notes below your transition, glide up two or three notes above it, then come back down. Slow enough that a whole step takes about two seconds. In the pitch studio you can watch the trace do this, and the trace is honest about the thing you cannot feel: where the line steps instead of sliding.

Do five, resting properly between them. Then move to "oo", which asks a little more. If the "oo" is going through cleanly, try "ee". Leave "ah" alone for now — the open vowel is the hardest crossing, and it is not where this block earns its keep.

Two rules for every siren in these two weeks.

First: never louder than conversation. If a neighbour through the wall would notice, you are too loud.

Second: if the glide breaks, do not restart it harder. Restart it quieter, slower, and a note lower. A break in the glide means the settings were wrong, not that you were insufficiently committed.

Slides are the same idea with edges. Take a five-note descending scale that starts above your transition and ends below it, sing it legato on "oo", and keep the top note light. Descending is deliberate. Coming down through the break from a light top is a gentler crossing than climbing into it, and the coordination carries over.

Warmups can do a lot of this work. Pick a pack with slides and sustained patterns rather than agility or staccato, then watch which notes in the lane fill in and which stay thin, and read the per-rep scores afterwards. It tends to be the same two or three semitones every pass. That band is the whole target of the block.

Vowel modification, plainly

Chapter 5 explained why the tract does what it does, and its modification ladder — which vowel leans toward which as you climb — is the reference to keep open beside you. Here is the practical version.

Going up through the transition, a vowel held in exactly the same shape gets harder to keep together. The fix is not to sing a different word. It is to let the vowel drift a small amount toward a more open, more neutral shape as the pitch rises. The listener still hears the word. You feel the space change.

Try it directly. Sustain a note two semitones below your transition on "ee" for four seconds. Then the same note, same breath, letting the "ee" widen slightly. Repeat a semitone higher, then again, walking up through the zone. Somewhere in there the unmodified version may start to feel pinched while the modified one does not. If that happens, the note it happens on is worth remembering.

The amount is smaller than you think. If you can hear yourself changing the word, you have gone too far. If a listener would call the diction muddy, too far. It is a nudge, not a substitution.

What it feels like when it starts working

Not like power. That is the surprise.

The crossing, when it goes well, feels like less rather than more. You arrive at the note above your transition and nothing happened — no gear change, no bracing, no held breath. Singers describe it in different ways: the sound getting narrower and cooler, the effort moving off the throat and into the middle of the body, or nothing much at all, which is why it is easy to miss. A clean crossing can pass unnoticed, because you were waiting for a sensation of achievement and got a sensation of ordinariness instead.

On the trace it looks like a slide that stopped having a scruffy patch in the middle. On a recording it sounds like one instrument rather than two. If you took a siren in week 5 and take the same siren in week 6, A/B them in the Recorder rather than trusting your memory of how week 5 felt. Memory is unreliable in exactly the direction that flatters recent effort.

Expect this on some days and not others, and expect the good days to be scattered rather than sequential. Two weeks is not long. The aim by the end of week 6 is a crossing you have felt often enough to recognise, not one you can produce on demand.

The days it does not work

There will be several. Here is the order to try things.

Go quieter. Half the volume you think is reasonable, then half again. Volume is the first thing to rule out.

Go lower. Move the whole exercise down two or three semitones and work under the transition instead of through it. A session spent below the break is not a wasted session.

Change the sound. If "ee" is fighting, go back to "ng". If "ng" is fighting, go back to breath work and gentle onsets, and leave the transition for that day.

Check the obvious. Chapter 12 lists the things that make a session read worse than your voice actually is — time of day, a cold coming on, a noisy room, headphone bleed. A voice that was fine on Tuesday and hopeless on Thursday is usually reporting on Thursday.

And if two or three sessions in a row are bad, stop trying to fix the block and repeat it. Spending three weeks here instead of two is a legitimate move, not a demotion. The transition responds to accumulated quiet repetition and to very little else.

One thing that is not a bad day: pain. Soreness, a raw feeling, or hoarseness that outlasts the session is a reason to stop singing, not something to practise through or work around. Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more.

The retest at the end of week 6

Block two closes with the range test. Run it on the last day of week 6, warmed up, under the conditions chapter 12 sets out — the same ones you used in week 1, or the two results are not comparable. Chapter 8's rule still governs it: only notes you would let another person hear count.

It gives you a low note, a high note and a voice type, and nothing else — not which notes are comfortable, and nothing about the crossing you have spent two weeks on. Two weeks of quiet siren work may move the edges a little or not at all, and both are ordinary. Log it, then leave the test alone until the start of week 12.

Try this

In the pitch studio, sing a slow "ng" siren from three notes below your transition to three notes above it, at conversation volume, and watch for the point where the trace stops sliding and steps. Do five. If any of them break, start the next one a semitone lower and quieter rather than trying again.

Weeks 7 and 8: extending the top

You have spent two weeks negotiating the transition rather than avoiding it. Now the work moves above it. This is the block people look forward to, and the one that most often goes wrong.

Nothing in these two weeks is about being loud, and nothing in them is about your highest note. The aim is to make the notes just above your transition ordinary — repeatable on a Tuesday, available at low volume, reachable without a wind-up. Whether the top of your range moves at all in fourteen days is not the point and is not under your control. How you get up there is.

Height and weight are two different dials

Most singers, asked to go higher, do one thing: they bring more of what worked lower up with them. More volume, more of the thick chest-dominant quality that felt strong in the middle, more push from the body. That is adding weight, and above the transition it works against you. The folds need to thin and lengthen as pitch rises. Carrying the heavier setup upward asks the mechanism to do two contradictory things at once, and that argument resolves as effort, sharpness, or a crack.

So treat them as separate dials. Going up should feel like the sound gets narrower and lighter, not bigger and harder. If the note above needs a bigger push than the note below, you are moving the wrong dial.

The practical version is quiet practice. For these two weeks, everything above your transition happens at a volume you could hold a conversation over. Not a whisper — a supported, quiet tone with the breath steady underneath it, in the sense chapter 4 gave the word. Quiet strips out the option of forcing. If you can only find a note by getting louder, you have not found the note. You have found a way to shout at it.

Head voice before belt

Establish the lighter mechanism above your transition first, over weeks, before you go near carrying weight up there.

Two reasons, and neither is aesthetic. Head voice teaches the coordination you actually need: thinner folds, a tract shaped to let the note out, breath that stays even rather than surging. Once that is reliable, adding a little body to it is an adjustment. Done the other way round, you spend months hauling a heavy setup at a wall.

The second reason is workload. Ten minutes of quiet head voice above the transition is a light session. Ten minutes of repeated belt attempts near your ceiling is a different proposition.

If you sing styles where belt is the sound — pop, rock, musical theatre — this is not a detour away from what you want. A reliable belt tends to sit on top of a well-organised head voice, and the mix chapter 2 described lives there too. Chapter 10 had you find the pitch where belting stops being available. For these two weeks, that pitch is a ceiling you do not test. Sing below it in your usual voice, and above it lightly. The region right at the ceiling, hammered over and over, is the fastest way to end a block sore.

What the two weeks look like

Keep the rhythm of the earlier blocks: three or four sessions a week, around twenty minutes, never two hard days back to back.

Start every session lower than you intend to finish. The middle-voice work from the earlier blocks is your on-ramp, not a warmup you have graduated from — five or six minutes of it in the Warmups packs before anything goes above the transition. A cold voice at the top of your range is where the trouble is.

Then the ascent work. Slides are still the best tool you have: an "ng" or a soft "oo" carried up through the transition and a few notes past it, slowly, at that conversational volume. From there, small descending patterns that start above the transition — a five-note scale down, a simple arpeggio from the top — because starting high and coming down lets you arrive in the light mechanism instead of climbing into it. Then the same patterns ascending. Vowels stay modified through the passaggio the way you worked on them in weeks 5 and 6. That does not stop being necessary once you are above it.

Add one or two semitones at a time, and only when the current top note is comfortable on three separate days. Not one good day. Three. Voices flatter you on a good day, and a semitone claimed on a Wednesday you felt great often will not be there on Saturday.

Finish every session below the transition. Two or three minutes of easy middle-voice humming or gentle descending slides costs you nothing and leaves the voice in better shape for tomorrow.

The limits, stated plainly

Ten minutes maximum above your transition in any session. That is total time up there, not total session time. Twenty minutes of ascent work is not twice as good; past a point you are practising fatigue.

Three attempts at any note near your top, then move on. Not five, not "one more". The fourth attempt at a note that failed three times is rarely better, and it is where strain accumulates.

Stop for the day the moment singing costs more than it did twenty minutes ago. If a note that was easy at the start of the session now needs a run-up, the session is over. That is not weakness. It is the correct reading of the signal.

No high-volume top notes this block. None. If you want to know what your loud top sounds like, that question belongs after twelve weeks, not inside week seven.

Why the top is where people get hurt

Two things stack up there. The mechanism is working at its finest margins — thinner, longer, faster, with less room for error — and the singer, wanting the note badly, is applying more force than anywhere else in the range. Fine coordination plus maximum effort is a poor combination in any physical skill, and singing is no exception. Add repetition, because a high note that nearly worked invites another go, and you have the standard recipe.

The other trap is that it does not hurt at the time. Adrenaline, the excitement of hearing the note come out, and a hot voice all mask fatigue while you are singing. The bill arrives the next morning, as a voice that will not start cleanly or a middle that feels thick and unresponsive.

Some tiredness after a session aimed at the top is ordinary. These are different, and they are reasons to put the session down rather than sing through: pain, hoarseness or breathiness the day after, notes in your middle that were fine last week and are now unreliable, or any sense of something catching. Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more.

Reading what happened

Record one take a week in the Recorder: a short ascending pattern through the transition to a comfortable top note, at the quiet volume. Two weeks later, A/B those takes against each other rather than against your baseline. You are not listening for a higher note. You are listening for whether the notes above the transition sound less startled — fewer scoops, less breath in the tone, a cleaner start.

If you have Coach, watch per-note accuracy in the band three or four semitones above your transition. Notes getting steadier there are the real result of this block, whatever the range test says. Your next retest comes at the start of week 12, the way chapter 19 sets out, and there is nothing to gain from checking your ceiling in the middle of a block aimed at it.

Some singers come out of these two weeks with a little more at the top. Some come out with the same top note, sung far better. The second is worth more, and it is the one that holds.

Try this

In the pitch studio, sing a five-note descending scale starting two semitones above your transition, at conversational volume, three times — and watch whether the trace is steady from the first note or only settles by the third. Arriving cleanly on the top note is this block's whole job.

Weeks 9 and 10: the bottom and the long phrase

After two weeks of adding height, this block goes the other way and slows down. The bottom of your range and the length of your phrases look like unrelated projects. They are not. Both get worse the harder you work at them, and both are governed by how steadily you spend air rather than how much of it you can move. Everything in these two weeks is a study in doing less on purpose.

The bottom is not the top upside down

Weeks 7 and 8 asked you to keep weight off the top. The low register asks for the opposite discipline, and most singers find it harder, because the instinct is so specific and so wrong: when a note is low, people reach down for it.

Reaching down has a recognisable set of symptoms. The jaw drops further than the vowel needs. The chin tucks. The larynx gets pressed downward to manufacture depth. The tone goes darker and woofier inside your own skull, and thinner on playback than the note a semitone above it. You are adding effort to a mechanism that sounds best when it is left mostly alone.

What is happening down there is the reverse of the top. The folds are shorter, thicker and slacker, and they vibrate in a way that does not want much pressure underneath. Push more air at a low note and it does not get bigger. It gets rough, or it gives out and drops into a rattle. Low notes are made by release.

Three practical consequences follow.

Low notes are quiet, and that is not a fault. They carry less energy and less of the brightness that makes a voice cut through a band. The commanding low notes you hear on records are shaped by mic placement and a mix as much as by the singer. Judge yours on tone and steadiness rather than volume, and know that leaning closer to your own mic flatters the bottom in a way that will not survive a room.

Approach from above, not from below. Attacking a low note cold, from silence, invites the reach. Start a few notes higher and come down to it on a slide or a scale, and it often arrives with no adjustment at all. That is why nearly all the low work in these two weeks descends.

Your test floor and your usable floor are different notes. The range test reports a low note, and it is the lowest one you would let another person hear — fry and scrapings do not count. Even so, it is rarely a note you can start cleanly, hold for four seconds, and sing a word on. A few semitones above it sits a note that behaves. That higher note is your real bottom for repertoire, and it is the one worth knowing.

What to do in the sessions

Keep the shape you have been running: three or four sessions a week, twenty minutes each. Do not stack low work on consecutive days at first. Twice a week is plenty in week 9.

Give roughly six minutes to descending warmups. Pick exercises from the Warmups packs that sit in your lower middle and move downward — five-note descents, slow slides, hums. Start each one above the bottom of your range and let it walk down. When it stops being easy, stop that repetition. There is no version of this where forcing one more semitone helps.

Use a closed, relaxed sound for most of it: a hum, an "uh" as in "cup", a soft "oh". Closed sounds keep the tone gathered without asking you to darken it by hand. If you catch yourself trying to sound low, you have already started pushing.

Watch the pitch trace on the descents. The low register tends to go flat before it goes rough. A trace that sags a few cents under the target on the way down, session after session, is usually telling you the note is under-supported rather than out of reach.

One caution, because the low register is deceptive: pressing the larynx down does not announce itself the way a strained high note does. It rarely stings. You can do it for twenty minutes and notice only afterwards that speaking feels heavy. Keep the low sessions short. Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more.

Phrases, not sustains

The other half of this block is length, and it starts with a distinction worth being fussy about.

The sustain test in Breath gives you a number of seconds on one steady tone. That number is a useful thing to track and a poor description of what a song asks of you. A real phrase changes pitch, changes vowel, has consonants that interrupt the airflow, and ends on a note you have to land rather than abandon. It costs more than a sustain of the same length, and it costs unevenly.

So the aim for weeks 9 and 10 is not a bigger number. It is finishing phrases with air still in reserve. The last stretch of a breath is where tone falls apart: pitch drifts flat, the larynx creeps up to compensate for pressure you no longer have, and the line ends in a squeeze.

If you keep arriving at the end of a line empty, the fix is rarely a bigger inhale. Look upstream. A breathy onset spills air before the note starts. Too much pressure spent in the first half leaves the second half paying for it. A breath taken late is not a real breath. And a breath point chosen where the words happen to leave a gap, rather than where the phrase actually divides, will cost you every time through.

Plan your breaths and keep them in the same places. A breath you decide on in advance is a musical event. A breath you take because you ran out is an accident, and it lands somewhere different each time, which makes the line unrepeatable.

For the breath half of each session, four or five minutes: the breath drills chapter fourteen describes, run with the exhale counts a beat or two longer than you used in weeks 1 and 2. Slow and quiet. They train an even outflow, not a large one.

Putting the two together

The last seven or eight minutes of each session is where these two weeks earn their place, and it is the first time in the program you work on real music rather than exercises.

Pick a song from Songs, transposed so its chorus sits in your middle rather than at either edge. Choose two phrases: the longest one, and one that dips low. Mark where you intend to breathe. Sing them at conversational volume, slowly if the tempo allows, and record every attempt.

Then listen. You are checking three things — whether the low notes stayed released or got manufactured, whether the ends of phrases held their pitch and tone, and whether you breathed where you planned to. A/B the take in the recorder against one from earlier in the week. Do not trust the memory of how it felt; it is unreliable in both directions.

With Pro, the coach report's per-note accuracy will point at which notes in the phrase are costing you. The bottom of a range often reads worse than the middle, which is what descending work is for. It is not a reason to go hunting for a new low note.

Run one sustain test a week, at the same time of day, and log it. There is no range retest in these two weeks; the next one comes at the start of week 12. Two weeks of honest low work may not move your low note at all. What it can change is whether the bottom you already have is a note you can sing a word on.

Try this

In Breath, run the sustain test once, then sing the longest phrase of your current song from Songs and see whether you finish with air left over. If you do not, plan a new breath point rather than trying to inhale more.

Weeks 11 and 12: putting it in a song

Ten weeks of exercises have been building coordinations in isolation: a steady middle, a slide that survives the transition, height without weight, a low note that stays relaxed, a phrase that lasts. None of it counts for much until it survives contact with a song, where the notes arrive in an order you did not choose, at a tempo you cannot slow down, while you are also thinking about words.

These two weeks are where that transfer happens. They are also where you find out what the previous ten did, because at the start of week 12 you retake the range test and record the same song you recorded in week one.

The work is still twenty minutes, three or four times a week. What changes is the ratio. Warmups shrink to the smallest set that gets you working — five minutes, the pack you have been using, nothing new. The rest is song.

Picking the song

One song. Not three, not a set. You are testing transfer, not building repertoire, and one song sung twelve times teaches more than four songs sung three times each.

Start with the aspiration song you wrote down in chapter 14. If it meets the two tests below, it is the obvious choice. If it sits well above your transition for a whole chorus, park it and pick something nearer; it is still the target, just not the song you test on.

Pick something you already know by heart. Learning notes and words while also applying new coordination gives you two problems that mask each other. If the melody is already in your ear, everything that goes wrong is a singing problem, and you can see it.

Pick something that mostly sits in your comfortable middle, with a few notes reaching into the zone you worked on in weeks 5 through 10. A song that lives entirely in easy territory tells you nothing. A song that sits above your transition for two minutes is not a test, it is an ordeal. Chapter 20 covers auditing a song properly; the rough check for now is where the chorus sits, because the chorus is where you spend the most time and where the tessitura bites.

If the song is in the wrong key, move it. Songs keeps melodies auto-transposed into your comfortable range, which does the arithmetic for you. Changing key is standard practice, and chapter 21 makes the case at length. What matters this week is that a song fighting you by two semitones can hide whatever changed in the last ten.

Record it once, early in week 11, before you have worked on it. That is your working copy — separate from the week-one baseline take, which you leave alone.

Week 11: taking it apart

Sing the whole thing once at the start of each session. Then stop singing it whole and work on pieces.

Find the hardest phrase. Not the highest note — the hardest phrase, which is usually the longest one, or the one that climbs into your transition on an awkward vowel, or the one that arrives right after a spot with no time to breathe. Most songs have two or three. The rest is comparatively easy.

Take that phrase into the pitch studio and treat it as an exercise. Sing it quietly. Sing it on a single vowel with no consonants, so you can hear what the line does without the words in the way. Then put one word back at a time. Watch the trace where it climbs. A phrase that reads clean in warmups and scruffy in context is telling you that the words, not the notes, are where the coordination falls apart.

Mark your breaths and keep them. Write them on a lyric sheet if that helps. Phrases often collapse because of a breath taken somewhere you never planned one, and you rarely notice taking it. The sustain work from weeks 9 and 10 carries over only if you decide in advance where the air is going.

Do not sing the song at full volume over and over. Two or three full-voice passes in a twenty-minute session is plenty; work quietly the rest of the time. Volume is what makes a song feel like a performance, and it is also what makes eleven repetitions of a chorus a bad idea. Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more.

Week 12: retest and compare

Do the retest at the start of week 12, not the end, so you have a few days left if it goes strangely.

Take the range test under the same conditions as week one, as near as you can manage: same time of day, same room, same distance from the mic, similar warmup. Chapter 8 covers taking it honestly and chapter 12 covers the ways a session can lie to you. Both are worth a reread before you press record, because a surprising retest result often turns out to be a measurement difference rather than a voice difference.

Then record the song. Same song, same key, same arrangement as your week-one take. Record it near the start of a session rather than the end, so fatigue is not part of the comparison.

Now A/B the two takes in the recorder. Listen straight through once, without stopping and without judging. Then listen again with a list.

Onsets first — the moment each phrase starts. Cleaner? Less scooping up to the note, fewer breathy leaks before the tone arrives? Onsets are worth checking early, because they are audible even to people who cannot describe what they are hearing.

Then the middle of the range, not the top. The notes you sing most often are where to look for steadier tone, less wobble on sustained notes, less of a gear change between one phrase and the next.

Then the transition. Not whether it is gone — expect it to still be there, since it is a feature of how the instrument works — but whether the approach is quieter, whether you brace less, whether the sound thins gradually instead of flipping.

Then the ends of phrases. Do they still have air behind them, or do they trail off and go flat on the last beat?

Reading it honestly

Two things go wrong here, in opposite directions.

The first is hearing nothing. Your own voice on playback is uncomfortable, and the discomfort flattens everything into "that sounds bad." If you cannot hear a difference, go to the numbers: the range history in Progress, the accuracy figures, and, if you have Coach, the per-note breakdown from week one against week twelve. Ten weeks is a short stretch on the timescale voices move, and chapter 11 explains why a trend needs more than two data points. Two readings are not a trend.

The second is hearing everything. A retest showing three extra semitones at the top after twelve weeks is more easily explained by a warmer voice, a bolder attempt, or a closer mic than by a longer range. Be suspicious of big changes at the outer edges. Be less suspicious of changes in the comfortable middle, in accuracy, or in how a phrase holds together, because those are harder to fake.

The honest summary is usually undramatic and specific: the middle is steadier, the onsets are cleaner, the passage that fell apart in week one survives at low volume but not at full volume. That is a real result. Write it down somewhere you will find it in three months.

Then find the sentence you wrote in chapter 14 about why you wanted to sing, and read it next to that summary. The numbers say what changed; the sentence says whether what changed is the thing you came for.

Choosing what comes next

Three sensible options, and the comparison tells you which.

If the transition work never quite took — if the retest and the recording both show the same disintegration around the same notes — go back to block two, weeks 5 and 6, and run it again. Repeating a block is not a setback. The transition is the slowest thing in the program to change, and it is a normal place to spend a second pass.

If the middle is solid and the top still frays, repeat weeks 7 and 8 with the same volume limits, and pair them with a second song that reaches slightly higher than the first.

If the whole thing held together, do not start another twelve weeks tomorrow. Keep three warmup sessions a week, add a second and third song, and retest the range in about a month rather than next week. Repertoire is the point of the exercises; sing some.

Try this

Open the recorder, put your week-one take next to your week-twelve take, and A/B the first four seconds of each — the very first phrase entry, nothing more. Onsets are the kind of detail you can hear without training, and four seconds beats sitting through both takes twice.

PART

Range and repertoire

Choosing songs that actually fit

Most songs that do not work were chosen in the first thirty seconds of hearing them. You liked it, you sang along in the car, the one big note came out, and you decided it was yours. Three weeks later the second chorus is a fight and you assume the problem is you.

Usually the problem is arithmetic. You compared one note in the song against one note in your voice, and neither of those numbers was the one that mattered.

The highest note is the wrong test

A song's highest note is a single event. It might last a beat and a half, once, in the bridge, on an open vowel, approached by step, with a band underneath covering everything. Notes like that are cheap. If it sits inside your performable range, it is not the thing that will beat you.

What costs you is the band of pitches the melody lives in — where it sits, breath after breath, for four minutes. That is the song's tessitura, and it is a completely different measurement from its ceiling. Two songs can share a top note and be nothing alike to sing. One touches it once and spends the rest of its life a sixth lower. The other treats it as the roof of a room it never leaves.

The second song fails in a specific way. Not on the high note. On the third phrase after it, when you have spent the reserve you needed for the rest of the verse, and your tone starts to spread and go breathy in a part of the range that was never hard before.

So: match the song's tessitura to yours. Check its ceiling second, as a veto, not as the criterion.

Where the melody actually lives

You do not need notation for this, or the ability to name pitches by ear. You need to find the melody's home.

Play the song and pay attention to which pitch it keeps returning to. Almost every melody has one — a note it leans on at the end of phrases, sits on for the long syllables, and orbits between moves. Hum it, then find it on the keyboard in the pitch studio. That single pitch tells you more about whether the song suits you than the highest note ever will.

Then find the band around it. Where does the melody bottom out in a normal verse, and where does it top out in a normal phrase, ignoring the one or two showpiece moments? For most songs that spread is about an octave. Lay that octave against the band you have found comfortable in practice over the earlier chapters — not against the two edge notes the range test reports, which mark limits rather than comfort. If the song's home band sits inside yours with room at both ends, the fit is working in your favour before you have sung a note. If it sits at or above the top of that comfortable band — even where every note is technically reachable — you have found a song that asks you to work for its entire length.

Do the same at the bottom. A melody that parks below your comfortable floor will not fight you, but it will vanish. Low notes lose carrying power fast, and a verse sung under your working floor tends to come out as a mumble that a microphone flatters and a room does not.

Why the chorus decides

Verses are written to sit low. They carry words, they want intelligibility, and they leave headroom for what comes next. Audit a song by singing the first verse and you have audited the easiest part of it.

The chorus is where the song commits. It is higher, usually louder, built to feel like release — and it happens three or four times. A phrase that is fine once is a different proposition on its fourth pass, after two verses and a bridge, with no rest between the last two.

Then there is the final chorus, which in a great many songs is not the same as the earlier ones. It goes up. Sometimes the whole thing lifts a tone or two. Sometimes the melody stays put and the singer takes it an octave higher. Sometimes it just repeats four more times with ad libs stacked on top. Whatever the mechanism, the last chorus tends to be the hardest ninety seconds in the song, and it arrives when you have the least left.

So audit the last chorus first.

The audit, in order

Ten minutes, before you commit to learning anything.

Find the home note. Hum the chorus, find that pitch on the keyboard, and note where it sits relative to your comfortable middle. If it is above your middle, everything that follows gets harder.

Sing the last chorus, twice, at a conversational volume. Not the recording's volume — yours, quiet. If it only works loud, it does not work. Loudness hides a bad fit in the practice room and stops hiding it on the night.

Count the returns. How many times does the melody go to its highest region? Once is a feature. Eight times is the real difficulty of the song, and it is the number nobody thinks to check.

Check the phrase lengths. Find the longest phrase with no obvious breath. A song can fit your range perfectly and still be out of reach because it asks for a sustained line you cannot currently carry. That is a different problem with a different fix, and it is worth knowing which one you have.

Check the top note in context. Not alone. How is it approached — by step or by leap? What vowel is it on? Do you sustain it or pass through it? A leap onto a closed vowel you have to hold is a much bigger ask than a stepwise arrival you leave immediately.

Check what happens after the hard part. Recovery matters more than the peak. A quiet verse after the big chorus is the song being forgiving. Another chorus stacked straight on top is not.

Then run the betting test from chapter seven. If the answer depends on the day, the song is at your limit rather than inside it.

What to do with a song that does not fit

Three real options, and only one bad one.

You can move it. Changing key is ordinary professional practice, and the next chapter is about doing it well, so no more here beyond this: a song that is a tone too high is not a song you cannot sing. The songs library will transpose a semitone at a time if you want to hear the difference before you decide.

You can shelve it. Some songs belong to voices built differently from yours, and coming back in a year is a legitimate answer. Songs do not expire.

You can change what you do with it. The melody you are copying is one singer's choice on one recording. Take a phrase down an octave, simplify an ad lib, drop an ornament you cannot place cleanly. Singers have always reshaped material to suit themselves; it is not a concession.

The bad option is the common one: keep the song where it is, keep the melody exactly as recorded, and try to grow into it by repetition. That is rehearsing strain, and strain gets learned as reliably as anything else. Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more.

Choose songs that fit the voice you have now, and you spend the performance sounding like yourself rather than like someone reaching. You can hear the difference on a playback long before anyone tells you about it.

Try this

Open the recorder and sing only the last chorus of a song you are considering — twice through, at conversational volume, saved as two takes. Put them side by side in A/B compare and listen for whether your tone thins or spreads on the second pass. That second pass, not the first, is what the song will actually sound like.

Transposition without shame

There is a belief, common among people who have never been paid to sing, that a song has a correct key and that singing it anywhere else is an admission. You could not manage the real one, so you moved it.

It is worth knowing where the original key came from before you feel bad about leaving it.

In most recorded popular music, the key was chosen for one particular larynx on one particular set of days. Sometimes it was chosen for the guitarist, because the riff needed open strings. Sometimes the demo got written on a keyboard in whatever key the writer's hands fell into, and nobody went back to it. Sometimes it moved up a couple of semitones late in the process because the chorus sounded more urgent when the singer had to reach for it.

None of that is a fact about the song. It is a fact about a room.

Meanwhile, the corners of music with the longest and strictest traditions treat key as a variable. Art song is published in high, medium and low editions, and nobody files the low edition under failure. Opera has a long history of transpositions made for particular singers, some of which stuck. Jazz standards are learned in your key as a matter of course — a horn player and a singer settle it early in the rehearsal, in the same flat tone they would use to settle tempo. Church music gets moved so the congregation can actually sing it.

Session and theatre singers hand out charts in their key without a flicker.

The idea that key is fixed is a listener's idea, formed by hearing one recording several hundred times. It was never a singer's idea.

What you are actually matching

The instinct, when a song is too high, is to work from the top note down. Find the highest note, work out how far above your ceiling it is, drop the whole thing by that much.

That is the wrong end to measure from, for the reason the repertoire chapter laid out: the chorus is where the song lives, and the chorus decides whether you survive it.

So the question is not "can I reach the peak in this key." It is "where does the bulk of this melody sit relative to my comfortable middle." Find the band the chorus occupies — usually a narrow one, often less than an octave — and move the song until that band sits inside the part of your voice that behaves when you are not thinking about it. Then check what happened to the top note. Then check what happened to the bottom.

That last check is the one people skip, and it is where most transpositions fail. A song is a container with two ends. Drop a high chorus by four semitones and you have dropped the verse by four semitones too, and the verse may have been sitting three notes above your floor to begin with. You solve a screaming chorus and inherit a verse the microphone can barely hear. If a song's total span from lowest note to highest is wider than the band you can actually use, no key exists that fits it — and it is better to find that out on a first pass than after a month of rehearsing it.

A practical way to do this: open the song in Songs and step it a semitone at a time with the transpose control, singing the chorus properly at each step rather than checking notes in isolation. The control moves a full octave in either direction, which is more room than you will need. If you have taken a range test, Fit to my range centres the song's notes inside your measured range and gives you somewhere to start — a first guess, not a verdict. Step either side of it and use your ears.

Two habits make this go faster. Try the key you think is right, then try one semitone above and one below, because the middle of three tests is more trustworthy than a single test. And sing the whole chorus, not the hook. The hook is fine in almost any key, which is exactly why it is a bad witness.

What changes when you move it

Transposition is not neutral. Something is always traded, and it is better to choose the trade than to discover it at a gig.

Where the song crosses your transition. This is the big one. A melody that sat comfortably below your passaggio in one key may straddle it in another, and a phrase that crosses back and forth over the seam is harder than a phrase sitting a little higher and staying put. Moving a song down does not always make it easier. Sometimes it drops the melody right onto the join.

Which words land on which notes. The vowel that felt open on a high note in the original key is now on a different pitch, where it may want a different shape. Consonant clusters that were manageable at the old height can get clumsy at the new one. Worth noticing, not worth agonising over.

The drama of the climax. Some songs are built so the peak note sits at the edge of the original singer's comfort, and you can hear the note costing something. Drop it five semitones and it becomes an easy mid-range note, correctly sung and inert. The answer is not to keep a key that hurts. It is to put the cost somewhere else — volume, timing, the breath you take before the phrase, how late you arrive on the vowel. A note you can sing gives you choices. A note you can barely reach gives you one.

The instruments. Guitar arrangements built on open strings become different arrangements in a key without them, which is what capos are for. Piano parts survive transposition better but change in feel. Backing tracks and band charts have to move with you, so settle your key before anyone learns anything.

The register the song sounds like it is in. A low key can make a bright song sound heavier and more intimate. A high key can make a warm song sound urgent. Neither is wrong. Just know you are changing the song's character, not only its height.

Where transposition stops helping

It has limits, and pretending otherwise is how singers end up in keys that suit nobody.

Past roughly three or four semitones in either direction, most arrangements start to notice. The bass lands in an unhelpful octave, the guitar voicings thin out, backing vocals collide with the lead. It is still doable, and a fifth or more is sometimes right, especially across voice types — but at that distance you are arranging, not transposing, and you should expect to make decisions about the instrumental parts too.

Downward has a floor that upward does not. Your bottom notes lose carrying power before they run out of pitch, and a chorus dropped into the basement reads on a recording as mumbling, however accurate it is. If you are transposing down and the take sounds correct but weak, you have probably gone one or two too far.

Harmony parts and duets do not transpose independently. If you sing with other people, the key is a group decision.

And the honest limit: transposition solves fit. It does not solve technique. If a song sits well in your middle and still falls apart, the key was never the problem, and moving it again will not help. The reverse is truer than most singers want to hear — grinding away for months at a key that does not fit your voice is not discipline, it is a longer route to the same conclusion. Sing it where you sound like yourself. Then, separately, do the work.

One more thing. If a key that felt fine starts to cost you, that is not a transposition problem and it is not something to solve on the page. Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more.

Otherwise: write down the key you settled on and stop relitigating it. Practising a song in a slightly different key every week is a good way to learn it slowly and perform it badly.

Try this

Open Songs, pick something that has always felt slightly out of reach, and use Fit to my range as a starting point. Sing the full chorus in that key, then step the transpose control up one semitone and sing it again, recording both takes in the Recorder. Line them up in A/B compare and listen for which one holds its tone on the last line of the chorus, not the first.

Learning from other singers' ranges

There is a way of reading another singer's range that quietly makes you a worse singer. You look up someone you admire, see a figure like E2–C7, compare it to your own, and file the difference away as the distance between you and them. Then you spend six months practising the wrong thing.

The famous ranges page holds 357 singers on one keyboard. Almost none of its usefulness lives in that first comparison.

What a cited range is, and how it got that way

A cited range is an aggregate. It is the lowest note anyone has found on any recording, paired with the highest note anyone has found on some other recording, and those two recordings may be twenty years apart. Nobody sat the singer down and ran a range test. Someone listened, transcribed, and posted a figure; someone else copied it; a third person found a lower note on a B-side and revised it down. That is the process. Where a source exists, the singer's page names it for the low note and the high note, and it is worth reading — "sustained through the second chorus of a studio ballad" and "one shriek at the end of a live take" are not the same kind of evidence.

Three things get flattened by a single low–high figure.

Mechanism. A cited span often stitches together notes made in completely different ways. A whistle-register top and a chest-voice bottom sit on the same bar as though they were two ends of one continuous instrument. They are not. A singer with a four-octave figure may have far less connected, usable voice than the number suggests, and the notes at either extreme may be unavailable to them in any actual song.

Studio. Records are edited, comped, and sometimes pitched. A note that exists on a record is not proof of a note that existed in a room. That is not an accusation about anybody. It is how records have been made for a long time.

Time. Voices change across a career. The figure treats thirty years as a snapshot.

None of this makes the library useless. It makes it a rough map, and a rough map is fine as long as you do not navigate by the decimal places. Treat every figure as approximate, and the extreme ends of every figure as the most approximate part of it.

The number you want is not on the chart

Earlier chapters separated the notes a singer can reach from the band their voice actually lives in. Cited figures are the first kind, collected under the loosest conditions of any measurement in this book.

What you want to know about a singer you admire is where the voice lives. Which fifth of the keyboard the choruses sit in. Where the melody spends its time when nobody is making a point. That is what tells you whether their songs are singable by you, and no reference page publishes it, this one included.

You can get at it yourself in about ten minutes. Pick three of their songs. Ignore the highest note in each one. Sing along quietly, or play the melody, and find the band of five or six notes the chorus keeps returning to. Write the three bands down. What you end up with describes what your voice would be asked to do for four minutes at a stretch, which a low-to-high span never does.

Now set that band against your own comfortable middle, which you find the same way rather than reading it off the range test — the test reports a low note, a high note and a voice-type label, and nothing about where a voice prefers to sit. Heavy overlap means their repertoire is worth your time whatever the extremes say. If your middle sits a fourth below theirs, every chorus will fight you even when the top note is reachable — and moving the key, which is normal and covered in the previous chapter, is the fix rather than the failure.

Using the library well

Filter to your own voice type first. The filter is a blunt instrument, and the label from your range test is a rough bucket rather than an identity. It still puts roughly similar instruments in front of you, which beats starting from a name you already had in mind. The genre hubs do the same job from the other direction.

Read the "You vs" panel for overlap, not for the gap. Once you have taken the range test, each singer's page draws your range against theirs and reports how much of their span yours covers. The part worth studying is where the two bars sit on top of each other. A high percentage suggests their material sits in territory you can reach.

A low one usually means the two of you live in different parts of the keyboard, which is information rather than a verdict.

Hear the range before you judge it. "Hear this range" plays the low note, glides up, and lands on the high one. Two octaves and three octaves look like different worlds on a chart and sound much closer than that. Getting the real distance into your ear does more against the mystique of a big number than reading about it will.

Visit the record-holders page once. It exists, it is fun, and it is the least instructive corner of the library. The page says so itself: the widest figures are the ones with the thinnest evidence behind them. Look, enjoy it, and set nothing on it as a target.

Look sideways, not up. The most useful singer in the library is rarely the one you already love. More often it is someone a genre over whose comfortable band matches yours almost note for note, whose songs never came up because they were not on your radar. That is what a searchable list of 357 voices is for — finding material that fits an instrument like yours, not confirming that a famous person could sing higher than you.

What imitation is good for

Copying another singer is one of the oldest ways to learn and one of the easier ways to end up straining. The difference is in what you copy.

Worth imitating: phrasing. Where they breathe. How they place a consonant against the beat. Whether they arrive on a note straight or slide into it. When they drop to almost nothing, how long they hold a note before letting vibrato in, which vowel they choose on a hard one. All of these are decisions. They are learnable, they transfer between voices, and stealing them is how style has always travelled.

Not worth imitating: timbre, weight, and effort. Timbre is mostly the shape of a vocal tract you do not have. The same choice, made in your body, produces your sound instead of theirs, and that is the correct result rather than a shortfall. Weight is worse. Matching the thickness of a bigger voice means carrying chest weight higher than your instrument wants, and a top built that way tends to arrive tight and tire early in a session. Effort is the trap of copying by sound alone: you hear a note that reads as enormous and reproduce it by pushing, when the original may have been a moderate sound in a good room with a compressor doing the rest.

A rule that holds up under pressure: imitate what a singer decided, not what they are made of. When you catch yourself straining to match a texture, you have crossed from the first list into the second, and the practice has stopped paying.

One signal ends a session rather than shaping it: if singing hurts, stop, because pain is not evidence of work getting done. Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more.

Try this

Open the singers library, filter to your own voice type, and open three singers you would never have called your kind of artist. On each page, look at the "You vs" panel and where the two bars sit on top of each other. That shared middle is where their songs would actually live in your voice.

Building a set you can survive

Every song you have chosen so far, you chose alone. You checked its tessitura, found a key that sat where you sing well, and learned it in a quiet room where you could stop whenever you liked. A set gives you none of that. A set is a chain, and each song hands the voice to the next one in whatever condition it left it. Song four is not sung by the voice you had at the start. It is sung by the voice that three earlier songs built or spent.

So the question changes. Not "can I sing this?" but "can I sing this after that, and still sing the two that come next?" The second question is the one an audience hears the answer to.

The set has an arc your voice can feel

Think of the whole night as one long phrase: a warm start, a middle where you have the most to give, a peak, a landing. You are not equally capable at all four points.

Early on, the voice tends to be stiff and unreliable at the edges. Some way in — assuming you have not spent it in the first ten minutes — you are usually in the best shape you will be in all night. Warm, responsive, the transition behaving, the top there without a fight. That window does not last. Later the voice gets heavier at the top, quiet singing gets harder before loud singing does, and pitch at the extremes starts to drift a little before you notice it consciously.

That shape suggests a placement rule: **put the hardest song in the second third of the set, not at the start and not at the very end.**

The instinct to open big is understandable. It is the song you are proudest of. But an opener is sung on a cold voice in an unfamiliar room, with adrenaline pushing you louder and heavier than you rehearsed. That combination is where singers get into trouble — not because the note is impossible, but because they arrive at it unprepared and compensate with force.

The instinct to close big is more defensible, because it makes a good ending. But it means spending the whole set knowing you have to be at your best at the moment you are least likely to be. If you want the demanding song last, the rest of the set has to be built around protecting it, which means most other songs give something up. Sometimes that is the right trade. Make it on purpose.

Recovery songs are real songs

The word "recovery" makes people picture filler. It should not. A recovery song is a good song that happens to be easy on you: its tessitura sits in the middle of where you sing most easily, it stays clear of your transition, it does not ask for sustained high notes, and it does not need volume to work.

Two or three of these across a set change how much voice you still have at the end. They do not have to be slow. Tempo is not the load. Tessitura, volume, and where the phrases sit relative to your break are the load. A brisk song sitting comfortably in your middle is more restful than a ballad that parks you a semitone under your passaggio for four minutes.

A useful test: sing the chorus of a candidate song three times through at performance volume. If you would happily do a fourth, it can serve as a recovery song. If the third already feels like work, it is a demanding song in a quiet costume.

Place one after your hardest song, always. Place another before it if the two songs leading in are both heavy. And be honest that songs shift categories — the recovery song at minute forty is not as restful as the same song at minute ten.

Keys across a set, not just within a song

You already know that transposing a song is normal practice. What gets missed is that the keys interact.

If four songs in a row put their choruses in the same handful of notes, you have written one very long song in one very narrow band. That band tires, and it is often the band around your transition, because a lot of popular writing sits there. The set feels harder than the sum of its parts and you cannot point at which song did it.

So spread the load. When you lay the set out, write down the tessitura of each chorus — the few notes each chorus actually lives in — and read down the column. You want variety there. If three cluster, move one by a tone or two, reorder so they are not adjacent, or swap one out.

Two more considerations that only show up in a set. If a song's chorus sits right at your break, it is a bad neighbour: put nothing demanding on either side of it. And avoid two songs in a row that both push your ceiling, even when each is fine alone. The second is sung on a voice that has already been to the top once, and the top is where tiredness shows first.

Plan for the room and the night, not the take

Everything above assumes rehearsal conditions. A room does not give you those.

The loudest variable is monitoring. If you cannot hear yourself, you sing louder — not as a decision, as a reflex — and it costs you volume control for every song that follows. Sort monitoring before anything else. If you get one thing right at soundcheck, make it hearing your own voice.

A dry room, a loud stage, air conditioning, an hour of talking to people beforehand: none of it is dramatic alone and all of it stacks. Assume the room takes something off the top of what you can do. Build the set with margin, not at the exact limit of what you managed once at home on a good day. Sleep and hydration are part of the plan too, and they are easier to protect the night before than the hour before.

Then there is the part singers forget. Talking between songs is voice use. It is usually talking over a room, at a pitch you never warmed up, without breath support. Four minutes of chat spread through a set is four minutes of unplanned singing sitting between the songs you planned carefully. Keep it short, keep it quieter than feels natural, and put the longer bit of talking after a demanding song rather than before one.

Plan for repetition too. A set is rarely sung once. Two nights running, or two sets in a night, changes the arithmetic — the demanding song that is fine on Friday can be why Saturday goes badly. If a second night is coming, take something out of Friday rather than hoping.

Pain, hoarseness that will not clear, or any blood is a doctor's question rather than a practice one — chapter six says more.

Build it, then test the joins

Write the running order down. Beside each song, note the chorus tessitura, whether it crosses your break, and whether it is demanding, neutral or restful. Then look at the shape. You are checking the same four things every time: hardest song in the middle third, no two demanding songs adjacent, restful songs spread rather than clumped at the end, no long stretch of choruses in one narrow band.

Then rehearse the joins, not just the songs. Sing three consecutive songs straight through with the real transitions between them, talking included. That is how you find out song six was fine in isolation and is not fine after song five. Nothing else reveals it. A song you have practised a hundred times alone has never once been sung in the condition it will be sung in.

If a join fails you have three moves: reorder, retranspose, replace. Try them in that order. Reordering costs nothing, retransposing costs a rehearsal, replacing costs a song. Most failed joins go away at the first.

And keep a spare — one song you can sing on any voice in any condition and would not be embarrassed to play, for the night the room is dry, the monitors are bad, and the plan has to change halfway through.

Try this

Open the Recorder and sing the choruses of your set, in order, back to back, as one take. Do it again at the end of a long practice session and put the two side by side in A/B compare. The chorus that falls off most between the fresh take and the tired one is the one to move, transpose, or protect.

A note on what this book is not

This is a book about practice, not medicine. Nothing in it diagnoses or treats anything. If singing hurts, if you are hoarse and it will not clear, or if you ever see blood, that is a question for a doctor or a speech-language pathologist. There is no waiting period, and you do not need to be sure it is serious before you go.